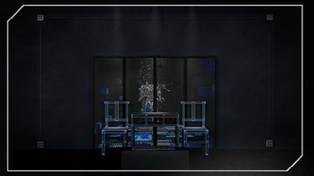
At the school theater, I often go to watch traditional Chinese opera performances, but I feel a sense of unfamiliarity——these movements make me feel like the actors are speaking "body language" that I don't understand. And there seems to be a thick layer of physical memory between the audience and the opera.

P 01 & P 02

The main research focuses on the severe challenges faced by the development of Chinese opera in contemporary society, as well as the lack of understanding and indifference of most people towards traditional Chinese opera.

P 03 & P 04

This section mainly studies the connotation and evolution of the core movements in Chinese opera, and transforms these movements into contemporary and codeable movements. Recoding is achieved through game interaction.

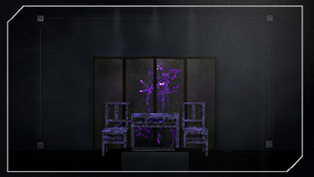


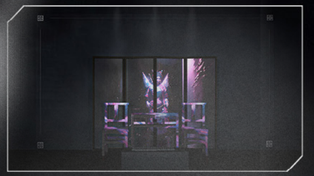
P 05 & P 06

This section mainly introduces the exhibition layout and game content design, a new medium of technology, creativity, and opera knowledge. Afterwards, digital presentations of various theatrical movements were presented.

P 07 & P 08

This section primarily outlines the development process and technical aspects of the Chinese Opera Movement Database. Finally, it presents the final presentation of the work, interview content, user feedback, and future development directions.





FROM SIGNAL TO CIVILIZATION

Movement — **Emotion**

Decode — **Re encode**

Design Description

Chinese opera is recognised as part of humanity's intangible cultural heritage and stands as an artistic treasure of the Chinese nation. However, the expressive forms, physicality, and emotional language of traditional theatrical movements have created a disconnect between contemporary practitioners and the general public. To bridge this gap and reshape audiences' understanding of this ancient art and its physical vocabulary, I have developed a Chinese opera movement database alongside an immersive experiential work.

RESHAPING EMOTIONAL CONNECTIONS FROM BODY MOVEMENT SEGMENTS

Video link : https://youtu.be/ZtS8rGtBefc

INSPIRATION

Incomprehensible Physical Movement Memory

At the school theater, I often go to watch traditional Chinese opera performances, but I feel a sense of unfamiliarity —— these movements make me feel like the actors are speaking "body language" that I don't understand. And there seems to be a thick layer of physical memory between the audience and the opera.

BACKGROUND

The Connotation Of Chinese Opera

Chinese opera not only generally forms the value system of Chinese culture, but also ultimately casts the poetic characteristics and personality charm of Chinese culture. Chinese opera is also the unity of **skill** and **spirit**.

SKILL

SPIRIT

The skills of Chinese opera are embodied in physical training and movement presentation

Stylized movement skills (posture, singing, reading, doing and playing) . Young people think it's not so important

Form muscle memory through repeated exercises: squat and roll, step up and fly, tiger jump, cloud hand, etc

Closed training system: "master apprentice system", strict child skill training, stylized action repetition, excluding non professional creators

The spiritual connotation of Chinese opera is embodied in behavioral ethics and spiritual influence

Spiritual inheritance of traditional Chinese Opera (ethics of etiquette, loyalty, filial piety, justice, good and evil causes)

The beauty and spiritual connotation of a program stem from its highly concise artistic form, which presents rich stories and meanings on stage

Transmitting normative behavior and moral standards through role setting: Who is a good person? Who can become an official? How do women behave?

Why Design

Misunderstandings of traditional culture ——— **Incorporating contemporary context**

Cultural interruption ——— **Establish a dialogue mechanism**

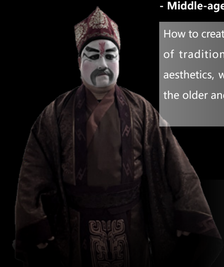
Lack of creative protection ——— **Adhere to principles and pursue innovation**

Vulnerability in contemporary systems ——— **Promote with contemporary media systems**

Design Goal

Current Sociation (For practitioners)

The development of Chinese opera faces severe challenges in contemporary society, with practitioners of different ages encountering unique predicaments:



- Middle-aged practitioners:

How to create works that not only retain the essence of tradition but also align with contemporary aesthetics, while simultaneously appealing to both the older and younger generations.

- Young practitioners:

The core issues of survival pressure, career prospects, and audience deficiency are the lack of industry attractiveness and sustainable development.

- Old practitioners:

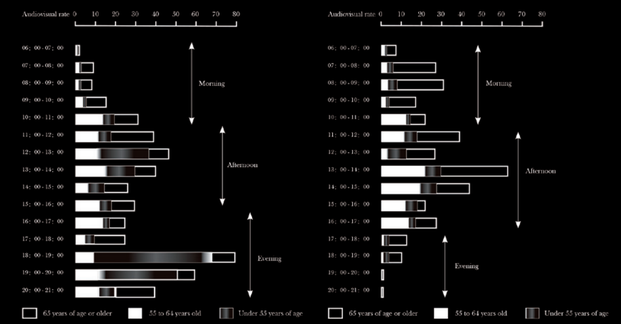
Due to physical limitations, the traditional method of "oral teaching" has limited efficiency. Dissatisfied with the current performers' skills and perseverance.

The predicaments of these three generations are intertwined, constituting an overall crisis facing the inheritance and development of Chinese opera. Solving these problems requires systematic efforts, as well as a renewed recognition and respect for the artistic value of opera from the entire society.

DESK INVESTIGATION

The Fault Between The Young Groups And The Opera System (For audience)

Viewership distribution statistics by age group for CCTV-11 Opera Channel (left) and Shanghai Opera Channel (right) -



The backbone of those who watch opera programs is the elderly. Most young audiences are also practitioners.

Young audience: Lack of interest and barriers to engagement, cultural gaps and aesthetic differences, stereotypes and cognitive biases.

Middle-aged audience: Due to the pressure of life and scarcity of time, cultural consumption does not prioritize traditional Chinese opera.

Old audience: Innovation in traditional opera seems hard for them to accept. Some changes to traditions are disrespectful to traditional opera.

Chinese Opera Genres Favoured by Different Age Groups -

What Forms These Physical And Emotional Barriers

The majority of people's indifference towards traditional Chinese opera primarily stems from a lack of understanding

It is difficult for people to understand opera movements

Dance coding that requires "special training" to understand

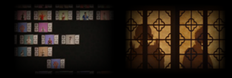
Contradiction between "skill and spirit" "integrity and innovation"

Young people believe that skills are not so important

The Way Of Chinese Opera Innovation Today

- Tobias Gremmler's work: Virtual Actors in Chinese Opera

- CCTV's Programme: The Theatre of Humanity



- Dans Digital's Production: Chang'an Revisited

Most cases focus on the visual expression of traditional Chinese opera, utilizing technology to create visual spectacles of Chinese opera.

The aesthetics of Chinese opera are also reflected in its cultural connotations, which means need updates in movement and emotion

How to repair faults? How to make modern audiences understand these ancient body languages?

DOCUMENT AND INTERVIEWS

The Core Stylized Movements And Distortion Timeline In Chinese Opera

Stylized Movement Name	Movement Behavior	Historical Connotation	Historical Sources	Expression Prevalent Today	Decoupling State	Cultural Transfer Status	Expression Forms Today
Wave Hands	Dance-like poses are formed through the coordinated movements of both hands and arms.	Express the spirit and demeanor of the character	The tenth form of the 24 Forms of Tai Chi	Yes	Not fully disconnect	From ritualistic and religious movements to aesthetic performances and daily exercises in life.	Fitness, pray, take a deep breath
Jump At Once	Stretching out both hands and jumping high with both feet off the ground, usually accompanied by other movements to complete the action.	Expressing emotions such as happiness and surprise is sometimes also used for avoidance	Life and combat scenes	Yes	Not fully disconnect	In contemporary combat, jumping height is no longer significant, and jumps are not emphasized for dramatic exaggeration.	Tilt backwards, Cover your mouth and nose, and clap your hands
Step Up Flying Foot / Flying Kick	Stand with feet in a T-stance, prepare, and then sequentially apply force to kick up, reaching a height at or above head level. This is often combined with arm movements to create a unified force.	Expressing battle scenes or the heroic and gallant temperament of characters, used to present the exciting parts of the drama	Wushu / Chinese kung fu	No	Basically completely disconnect	The heroic combat scenes that have lost their contemporary applications but still exist in many video games.	Hurdle, tackle, slam dunk
Tiger Jump	Imitate the posture of a tiger pouncing and roll or continuously roll while supporting the ground with hands.	Demonstrate explosive power and agility	Wushu / Chinese kung fu	No	Not fully disconnect	It exists in the trained crowd's ability to avoid or overcome obstacles.	Dive, pounce, slam the table and stand up, sprint
Squat And Turn Over	After squatting, with the right leg in front and the left leg stepping forward as the foundation, the upper body is bent over and leaned back to achieve a flip from right to left.	Initially, it was used to demonstrate balance and agility, but now it is more often used as a reminder action before important performances, serving both as a reminder and a transitional function.	Life scene	No	Basically completely disconnect	Having lost its contemporary application, this form of opening appearance is mostly simplified or transformed nowadays.	Clear throat/test microphone/look in the mirror/straighten collar/stretch

OCCUPATION + CULTURE + SYSTEM = MOVEMENT

The essence of movement is the symbiotic expression of profession, culture, and system. When these occuptions disappear in modern society, actions become "rootless".The movements in traditional Chinese opera have become a "cultural relic," and if not translated, they may be misunderstood or even discarded.

Survey methods and targets :

Literature review and summary, inquiry with opera actors, and replication and recording of their movements.Based on the connotation deduction and interviewing people, we can use an expression that is easy to understand.

Survey purpose and meaning :

Upholding principles and innovating needs to be conducted on the premise of respecting traditional culture, therefore, it is necessary to investigate the development process and connotation of opera movements

CONVERSION

Decoding and Recoding of Opera Actions

Chinese Opera movements are a bodily encoding of ritual, social identity, and history, and imbuing traditional movements with new contemporary meanings is a decoding and re encoding of Chinese opera movements.

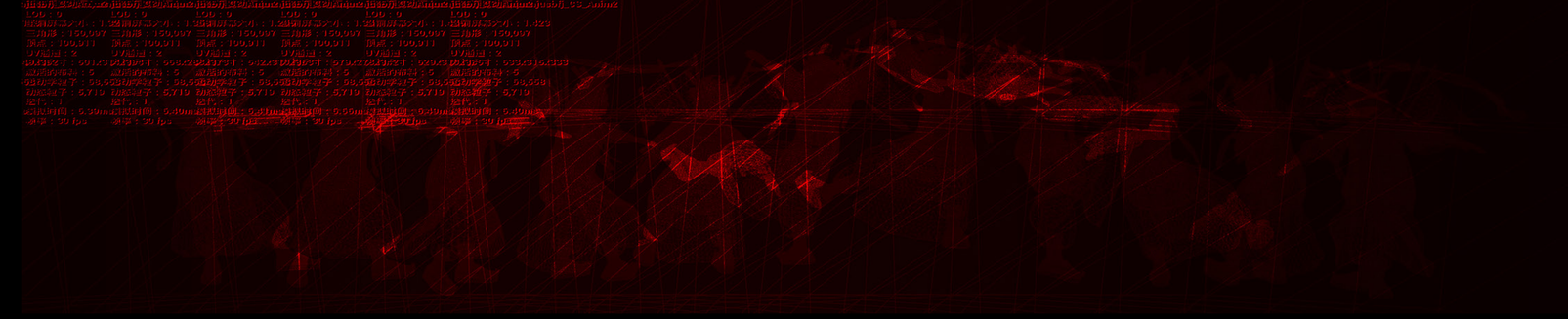
Movement Translation Database

The movement is an information system, but the "instruction manual" has been lost. When we try to understand it, we are actually building a neural connection bridge between contemporary people and traditional culture. Therefore, this project proposes the "Movement Translation Database".

Loss of movement connotation

Endow contemporary connotations

?



SQUAT

01

Wave hands

JUMP

02

Jump at once

SQUAT

03

Tiger jump

SQUAT

04

Step up flying foot / flying kick

SQUAT

05

Squat and turn over

WAVE SQUAT WAVE FLY JUMP TIGER JUMP SQUAT WAVE TIGER JUMP TIGER WAVE

Wave hands | Squat turn over | Wave hands | Step up flying foot | Tiger jump | Jump at once | Squat turn over | Wave hands | Step up flying foot | Jump at once | Tiger jump | Wave hands

Time line

The five basic movements of Chinese opera can be arranged in different opera performances through a timeline, and the modern meanings of these basic movements will also be presented.

DESIGN CONCEPT

The Combination Practice Of Space, Movement, And Emotion

This work want to provide a way for the fragmented public to experience the re encoding of the Dao (ritual system /social identity / history) through the translation of the opera movemnet system, in the absence of the opera movement (technique) "instrution manual". And I want to provide young people with a new technology and creative medium, as well as respect the spirit of opera, so that the older generation can understand and accept it.

Four scenes and characters

1、Traditional
2、Future
3、Fantasy
4、AIGC

Translation of 5 movements

1、Wave hands
2、Jump at once
3、Step up flying foot / flying kick
4、Tiger jump
5、Squat and turn over

5 types of sound effects

1、Rhythm drum beats
2、Strike a gong
3、Strike a big gong
4、Strike the drumboard
5、Vertical bamboo flute

Spotlight

Spotlight is the only light source used to illuminate the stage area and controller.

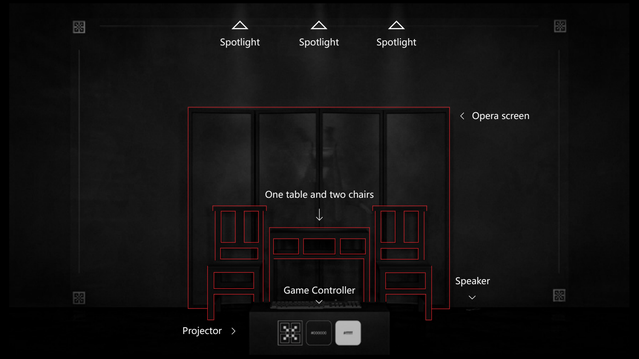
Physical equipment

Opera screen，One table and two chairs，Projection，Game Controller，Speaker

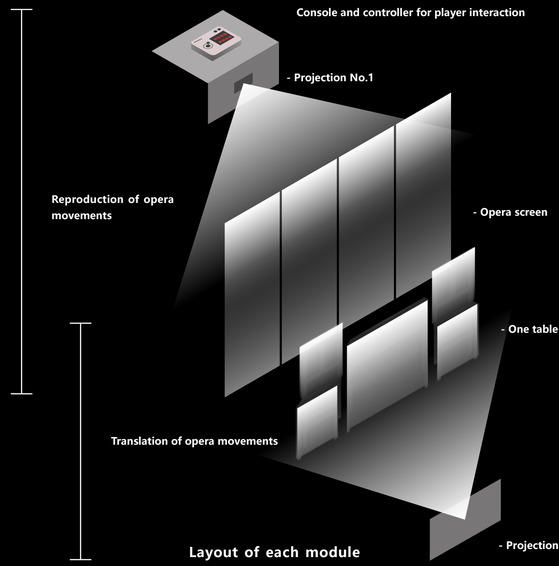
Interaction boundary

The controllable range of the controller，Air wall and ground

Exhibition plan



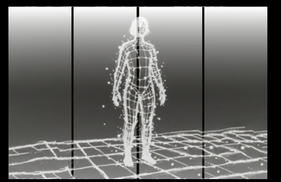
Exhibition Design



Layout of each module during the exhibition

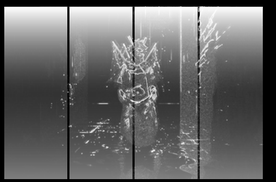
Prototype, Control And Music

Traditional scene ♪ - Traditional background music



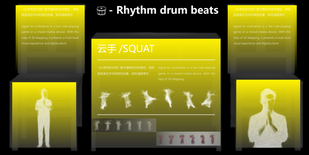
Future scene ♪ - Electronic background music

Fantasy scene ♪ - Fantasy background music



AIGC scene ♪ - AI background music

Game joystick controls the actions of opera characters, simulating the way arcade action games control characters, aiming to enhance players' sense of control and immersive experience.



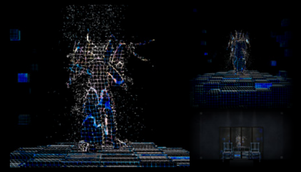
GAME DEMONSTRATION

Tangible Interactive Game System Flow Of Movements In Chinese Opera

Four scenes and characters design

- 01

- Traditional scene



- 03

- Future scene

- 02

- Fantasy scene

- 04

- AIGC scene

01

Wave hands

02

Jump at once

03

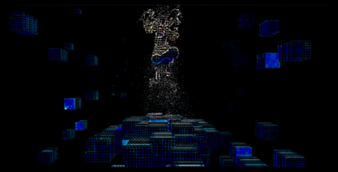
Step up flying foot / flying kick

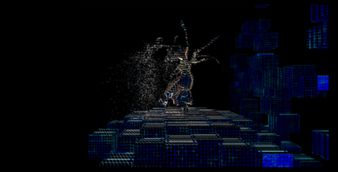
04

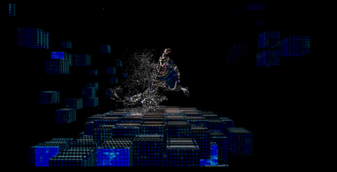
Tiger jump

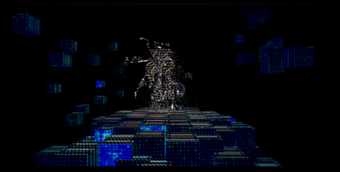
05

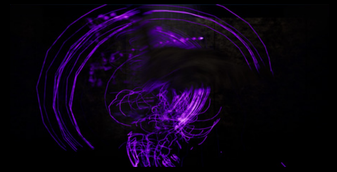
Squat and turn over

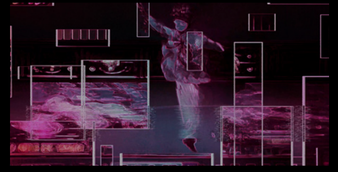












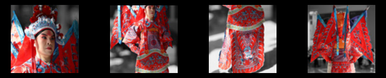
PRODUCTION PROCESS

Techniques And Software

Ps — BLENDER — Houdini — C4D — UE5

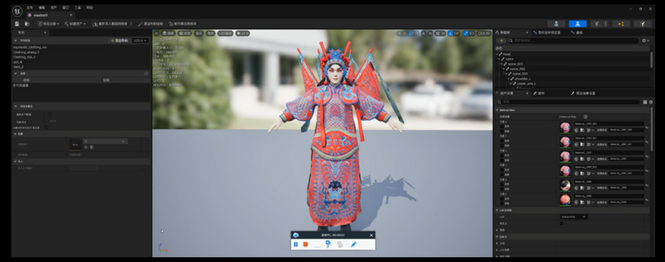
Motion Capture

Character prototypes and sketches



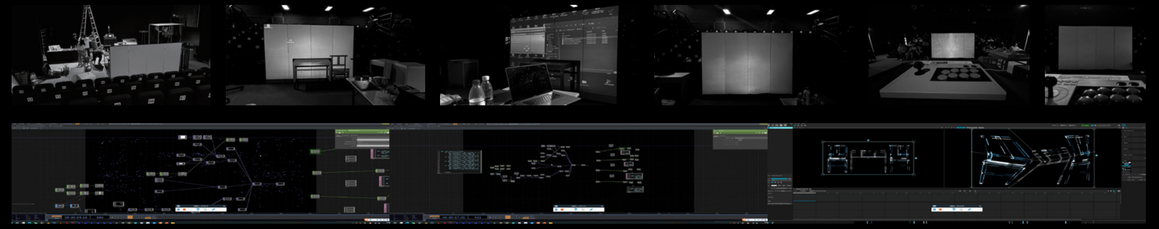
Preliminary establishment of rough mold by Blender

Bone Mesh Volume Import And Chaos Cloth Simulation

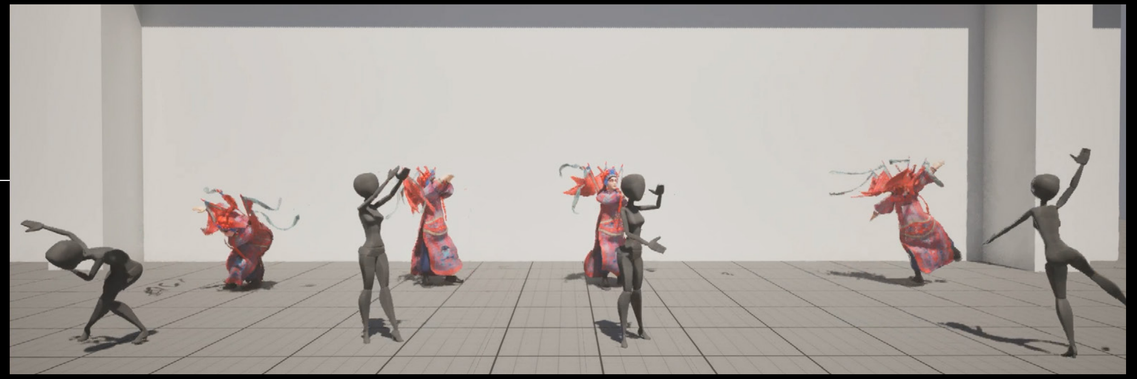


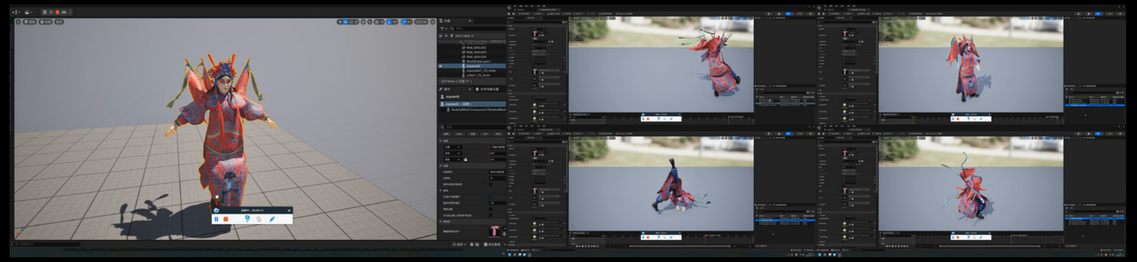
Processing of motion capture data and recording

3D mapping and scene layout

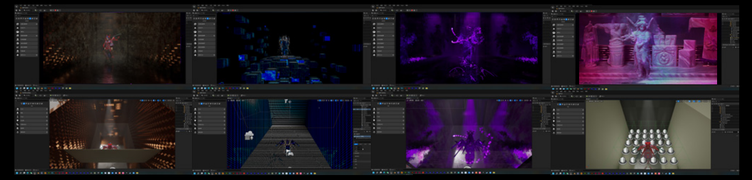


Character animation montage and animation state machine

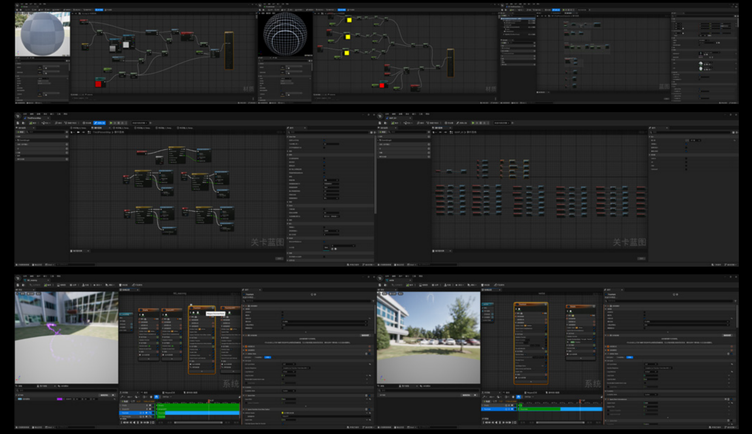




Character control and world blueprint



Character control and interaction blueprint



EXHIBITION

Real Playtime demonstration and Project Promotion

Location： Chaoyang Cultural Center，Nine theater

Time： 2024 Graduation Season 5.20 - 6.3

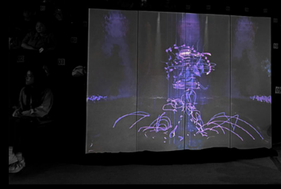
Exhibition： National Academy of Chinese Theatre Arts Graduation Exhibition

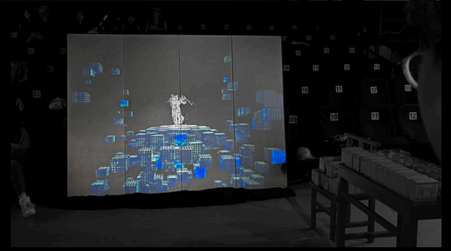
National Academy of Chinese Theatre Arts collaborates with Chaoyang Cultural Center to participate in Beijing Cultural Promotion Week and hold an exhibition. Over 20000 citizens attended the exhibition on-site.

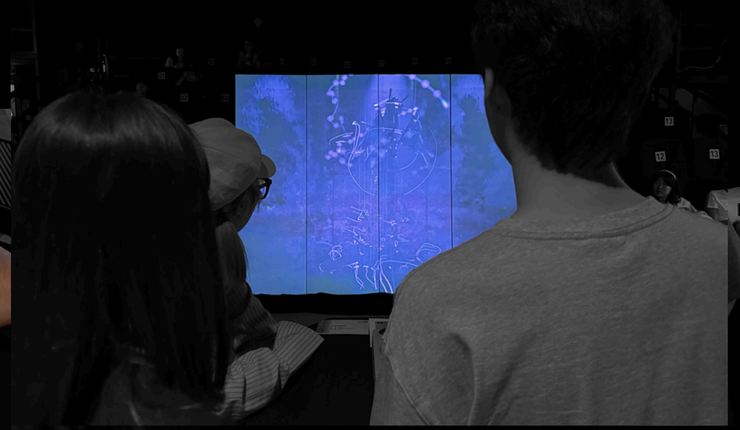
Internet display

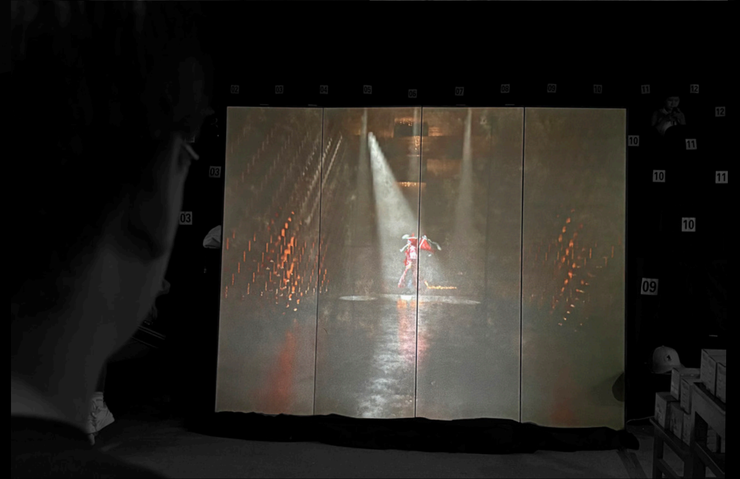
National Academy of Chinese Theatre Arts and China Stage Arts Association jointly released excellent graduation design exhibition on the Internet.

The exhibition held in a small cinema within the nine theater, where visitors can engage in immersive interaction or watch from the seats of the cinema.









Feedback And Suggestions

1 Being able to learn a lot of opera knowledge and have a strong sense of immersion.

2 This digital encoding method for opera movements can indeed assist in opera creation.

3 More opera movements can be added to increase richness. Multiple styles of traditional Chinese opera can also be added.

Future Prospects

In future, more Chinese opera movements and knowledge will be added to the gene pool, such as movements from Kunqu opera.

FIREWORK RUINS

Page 1 & Page 2

Mainly analyzed the causes of the appearance of fireworks relics in ecologically fragile areas such as grasslands, and studied various manifestations of this significant geological feature in the Anthropocene.

Page 3 & Page 4

Preliminary research has been conducted on the relationship between fireworks remnants and the setting off parties, areas, production factories, and environment. Afterwards, the field survey results were presented in a visual manner, and different fireworks and hazards were analyzed.

Project Brief

Firework ruins refers to the uncollected litter left behind after fireworks displays, particularly prevalent around festive periods and near specific scenic areas. Given the multiple harms I observed from fireworks discharge to grasslands, alongside structural damage to ecosystems, Through material testing, product development, and the creation of immersive digital content and associated systems, I have analysed this issue using primary research methods such as field surveys. This project aims to enhance public environmental awareness and action.

Page 5

By designing a series of methods, fireworks ruins can also be transformed into useful materials locally and can become materials to prevent grassland degradation. In addition, organizing activities to involve people together.

Page 6

This page mainly provides a detailed introduction to the digital twin islands and postcards, allowing participants to see the visual presentation of their labor achievements, such as the reduction and restoration of fireworks ruins on the islands.

BACKGROUND

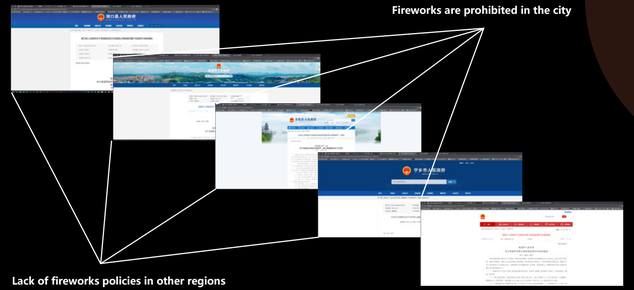
The Inspiration Of Fireworks Ruins

Fireworks ruins refer to the unattended garbage left behind after fireworks are set off. This is particularly common before and after festivals or around specific scenic spots. Due to fireworks bans in most cities, there are often more and more serious hazards caused by fireworks in rural and grassland areas.

Why Design	Design Goal
No one cleans up the grassland	Design a public benefit activity
Lack of recycling plan	Design a simple recycling system
Grassland ecological environment is fragile	Research on Prohibiting Types of Fireworks
People tend to overlook	Raise people's environmental awareness

CURRENT SOCIATION

Regulations



Fireworks are prohibited in the city

Lack of fireworks policies in other regions

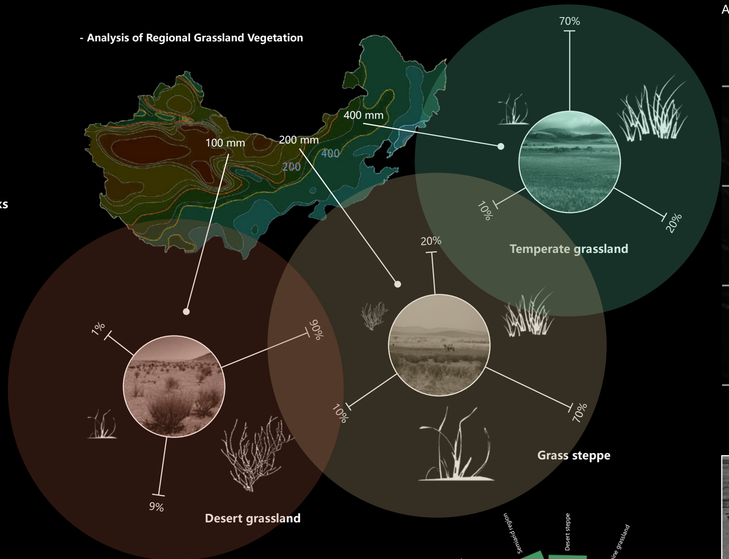
The ban on fireworks and firecrackers refers to the strict control and management measures implemented by relevant Chinese departments through legislation in the field of fireworks and firecrackers industry. There are 138 cities in China that prohibit fireworks. Nowadays, many places have changed their bans to restrictions, but fireworks cannot be set off in the main urban areas.

PRELIMINARY INVESTIGATION

An Unsuitable Place For Fireworks

In northwest China, there are different types of grasslands with varying precipitation distributions, and the ecological environment is relatively fragile.

- Analysis of Regional Grassland Vegetation



The damage of fireworks to different grasslands -

Arid region

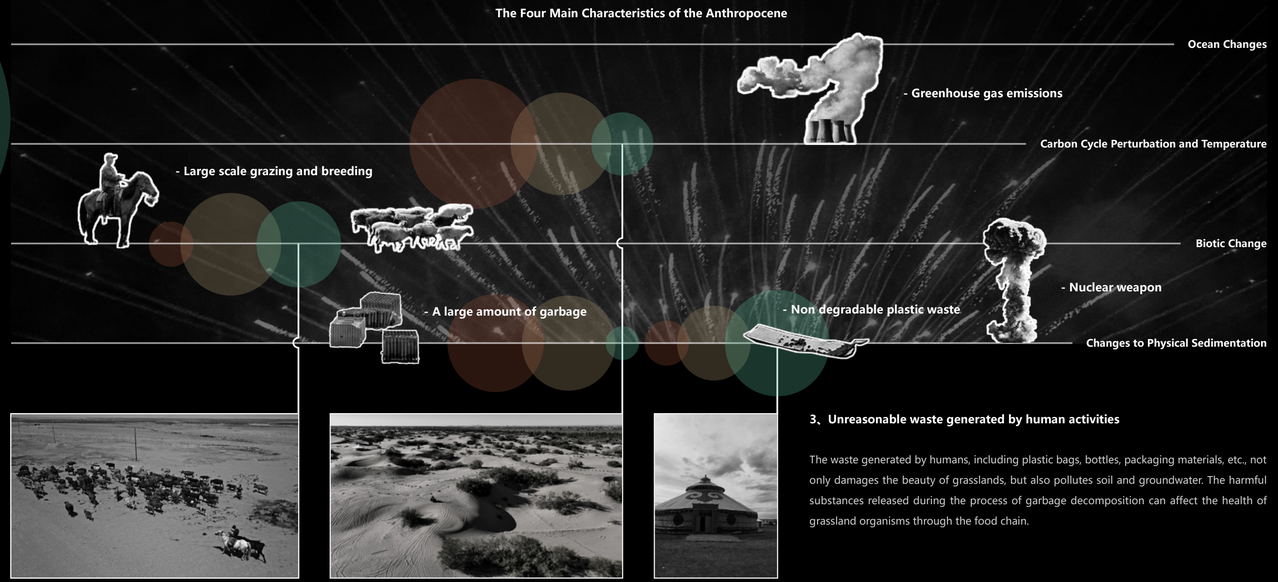
The ecological environment in semi-arid areas is extremely fragile, and the natural degradation of fireworks ruins here is relatively slow.

Semiarid region Semi humid region

These grasslands will attract more tourists, and the frequency of setting off fireworks is very high, causing significant damage to the grasslands.

Grassland Degradation And Governance

The problem of grassland degradation in China is the result of the combined effects of long-term natural conditions and human activities. According to data from the National Forestry and Grassland Administration, the ecological status of China's grasslands has shifted from "overall deterioration" at the beginning of this century to "overall improvement". The government has spent a lot of money to improve the long-term degradation problem, but about 70% of the grasslands still suffer from varying degrees of degradation. And there are still destructive activities like setting off fireworks in grassland.



1、Overgrazing and overcultivation

Human activities in animal husbandry have altered the natural balance, making it impossible for grasslands to be restored on their own. One third of the world's pastures are severely degraded due to overgrazing, and overgrazing is the main cause of land degradation.

2、Sand blown by the wind

Mobile sand dunes are one of the types of sand dunes classified based on their mobility. Its characteristics are: sparse surface vegetation, easy to move downwind under the action of wind, which poses a great threat to transportation and industrial and agricultural construction.

3、Unreasonable waste generated by human activities

The waste generated by humans, including plastic bags, bottles, packaging materials, etc., not only damages the beauty of grasslands, but also pollutes soil and groundwater. The harmful substances released during the process of garbage decomposition can affect the health of grassland organisms through the food chain.

The Reversal Strategy Of The Anthropocene

The Anthropocene refers to the most recent history of the Earth, and there is no exact starting year for the Anthropocene, but its significance can be controlled by humans. The geological changes in the Anthropocene can also be improved or even reversed through rational actions. So for fireworks relics, humans should strive to reduce their existence and minimize their harm.

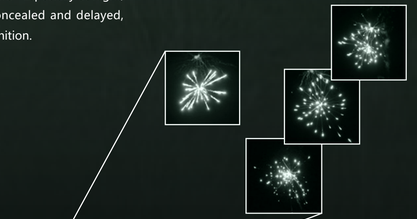
DESK INVESTIGATION

Tourists And Consumers

- tourists and Consumers

- Pay attention to fireworks and ignore the harm they bring

Human sensitivity to "instant happiness" is much higher than "potential risk", and this psychological tendency makes people more easily attracted by the beauty in front of them, and automatically ignore "potential problems in the future". Especially at night, the "harm" of fireworks is concealed and delayed, and is easily weakened by cognition.



- Let off fireworks

- Open ground

- Leave behind fireworks ruins

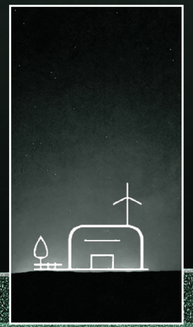
Inappropriate Places

- Grassland and pasture

- The appearance is good, but the location is not suitable

People like to set off fireworks on the grassland, and broadening their horizons can indeed enhance the viewing experience. However, the deeper reason is that the natural atmosphere of the grassland and the brilliance of fireworks form a unique emotional resonance, and the open space also provides convenience for event organization. However, this preference needs to be based on "respecting nature and safety regulations".

- Open ground and good view



Unqualified Factories And Sellers

- Unqualified factory

- Unqualified production causes more pollution

The pollution caused by unqualified fireworks is not limited to the environment, but also directly threatens human health.The emission pollution of unqualified fireworks is much higher than that of qualified products, mainly reflected in their doubled pollutant emissions and more complex composition :

1. Higher emissions of pollutants
2. More severe heavy metal residues
3. Increased unburned residue

Unqualified mining

Polluted water source

- Unqualified raw materials

- Unqualified production

- Unqualified emissions

Destroying The Natural Environment

- Natural environment

- The grassland ecological environment is fragile

The fragility of grassland ecological environment is the result of the combined effects of natural attributes and external disturbances.

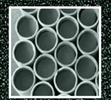
The core reason is the weak self-regulation ability of the ecosystem, low anti-interference threshold, and extremely difficult restoration after destruction.

- Industrial waste

- Industrial waste

- Industrial waste

The Relationship between Fireworks Industry and Environment



Unqualified factories cause more pollution

DOCUMENT AND FIELD RESEARCH

The Harm Of Fireworks To Vegetation

The setting off of fireworks and the resulting fireworks remnants (fireworks ruins) can pose various hazards to grasslands. Classification, quantification, and visualization are the main contents of this section.

Mineral Type Fluorescent Powder Labeling

In order to visualize the fireworks ruins, "fluorescent powder water composite solid dry particles" were applied to the fireworks ruins to associate them with the fireworks display.

Excitation By UV Lamp Irradiation

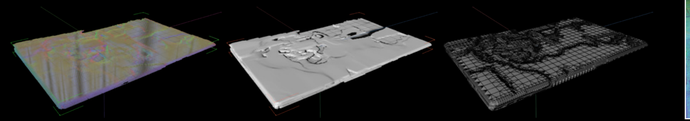
After application, irradiate the fireworks ruins with a UV lamp for 15 minutes, and different types of damage will appear in different colors. At night, people can clearly see the colored fireworks ruins.

Chemical Pollution

Fireworks contain various heavy metals, such as barium, strontium, copper, aluminum, magnesium, lead, arsenic, etc. Heavy metals are absorbed by plant roots and accumulate in plant bodies.

Heavy metals can interfere with plant metabolism: disrupting enzyme systems, affecting photosynthesis, respiration, nutrient absorption, and inhibiting root growth.

These metal salts can enter the food chain: heavy metals accumulated in plants may enter the food chain through herbivores, ultimately threatening human health.



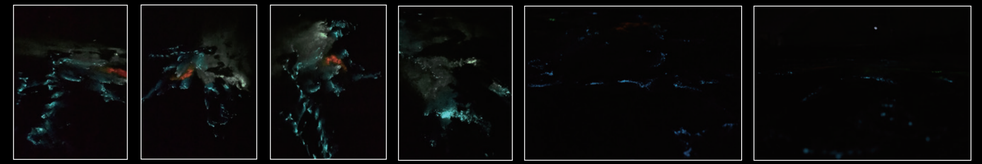
Physical Damage

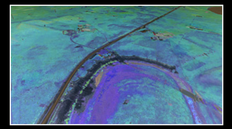
Thermal burns and physical impacts:

The high-temperature particles (sparks, unburned gunpowder particles) generated during the explosion of fireworks can cause open flames or local burns if they directly fall on plants (especially dry leaves, grass, dead branches, and fallen leaves), burning leaves, tender shoots, bark, and even the entire plant.

When larger fragments such as fireworks shells, cardboard tubes, and unexploded firework shells fall from a high altitude, they may directly break tree branches, injure or even kill seedlings or low plants.

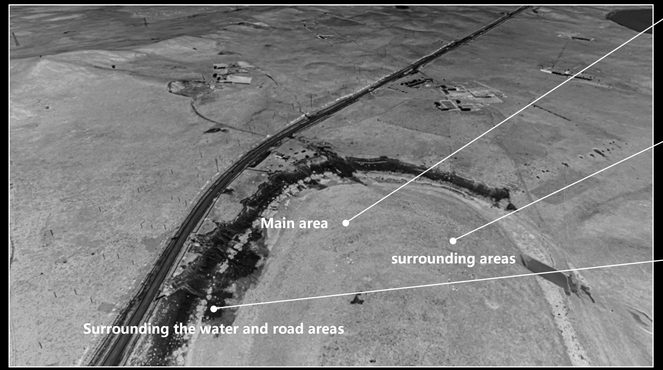
Visualization Analysis

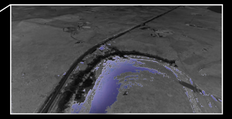


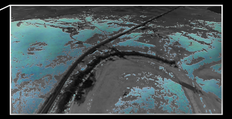


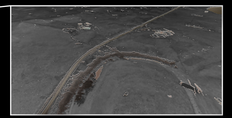
Damage type

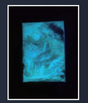
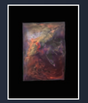
Fireworks classification









	Aerial	Firecrackers	Others	Dye
Direct damage				
Fragment				
Visible garbage				

Suggestions of different firework ruins

Fireworks that directly damage the grassland should not be set off :

After the fireworks explode, a large amount of high-temperature sparks and unburned gunpowder particles will scatter around, with a range of tens or even hundreds of meters, covering a wide range and easily igniting dry grass at long distances.

Fireworks fragments are difficult to handle :

These fragments of garbage can damage the microbial community in the soil, reduce soil fertility, and alter soil pH after sedimentation, inhibiting grass seed germination and root growth.And fireworks fragments are difficult to handle and collect.

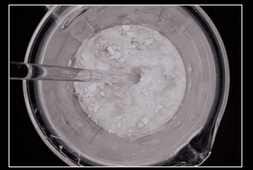
Can be collected, classified, and processed :

These large wastes can be recycled and reused, but harmful components and plastic waste need to be removed from them.
So the focus in the future will be on collecting and processing such fireworks ruins.

Color developer

Fluorescent powder water composite solid drying particles:

Combining fluorescent powder with water to form dry solid particles can better adhere and label the substance. And considering the biological toxicity, mineral type was chosen instead of chemical fluorescent solution.



DESIGN CONCEPT

Collection，Classification And Disposal Of Fireworks Ruins

By designing a series of methods, fireworks ruins can also be transformed into useful materials locally and can become materials to prevent grassland degradation.

Fireworks plastic waste

Plastic waste

The plastic waste in the fireworks ruins belongs to **non recyclable waste** that needs to be collected and **transported out of the grassland**.

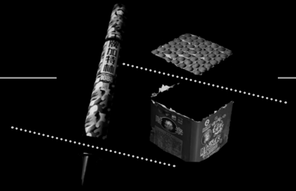
Remove plastic

The plastic waste mixed in the fireworks ruins needs to be **separated and collected** separately by participants in public welfare activities.

Soak in water

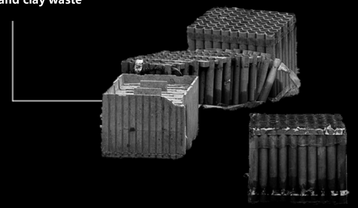
Long term soaking treatment can **render residual gunpowder** in fireworks remnants ineffective and **release packaging ink**.

Mixed garbage



Remove plastic

Paper and clay waste



Crush into three sizes

Crusher

Soak in water for 2 weeks

01 DAY 02 DAY

Add water

08 DAY 09 DAY

Drain off the liquid

03 DAY

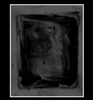
10 DAY

04 DAY 05 DAY

Add ink remover

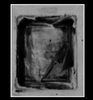
11 DAY 12 DAY

Clear the bucket



06 DAY

13 DAY



01 WEEK

02 WEEK

What Can Fireworks Ruins Be Used For

The processed fireworks ruins can be transformed in different environments, namely *a material for grass squares, a supplementary broadcast mixture and a raw material to increase soil organic matter.*

1、As a material for grass squares

Using locally sourced materials from fireworks ruins in grassland to further reduce costs

Reduce near surface wind speed in arid areas, prevent sand movement, and thus curb desert expansion

2、As supplementary broadcast mixture

Mixing with replanted grass seeds to improve soil fertility and grass seed survival rate

To curb the deterioration of grassland ecology and ensure the stable development of animal husbandry

3、Increase soil organic matter

Mixing biodegradable organic waste with livestock manure for composting treatment

"Grass square grid sand barrier" is an ecological protection engineering technology widely used in arid and semi-arid areas, and is the **most economical and effective** means of protection.

The "replanting" in grassland areas refers to the **artificial sowing** of high-quality grass seeds on degraded, desertified, or low vegetation covered grasslands to **enhance grassland ecology**.

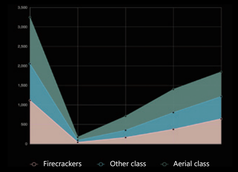
Transforming natural organic waste from grassland areas into **organic rich fertilizers** through microbial decomposition.

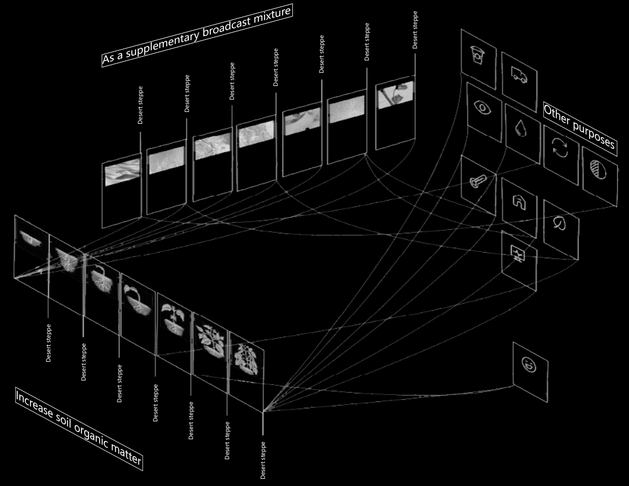
Public Benefit Activitie In Ulanqab

The number of fireworks ruins is highest near scenic spots or Mongolian yurts, so the collection of fireworks relics also begins in these locations on the Ulanqab grassland.

Quantitative analysis of fireworks ruins collection -

The collection of fireworks ruins can significantly reduce the amount of fireworks waste remaining on grasslands. **Some participants are also the ones who set off fireworks, and the event reminds them to collect the fireworks they set off.** Finally, useful materials will be used for protecting grasslands and ecological restoration.





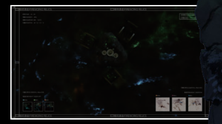
- Photos of activity participants

Participants who participate in the collection of fireworks ruins can receive postcards and enter the digital twin fireworks ruins island by scanning the code.

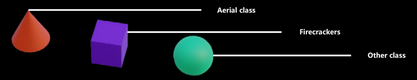
Design Extension

The Digital Twin Floating Island

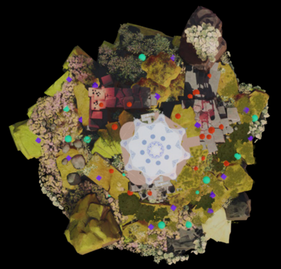
Participants can scan the QR code on the postcard to enter the digital island, which allows them to see the visual presentation of their labor achievements, such as the reduction and restoration of fireworks ruins on the island.



The fireworks ruins scene from the previous version has now been changed to an empty island that changes with interaction.

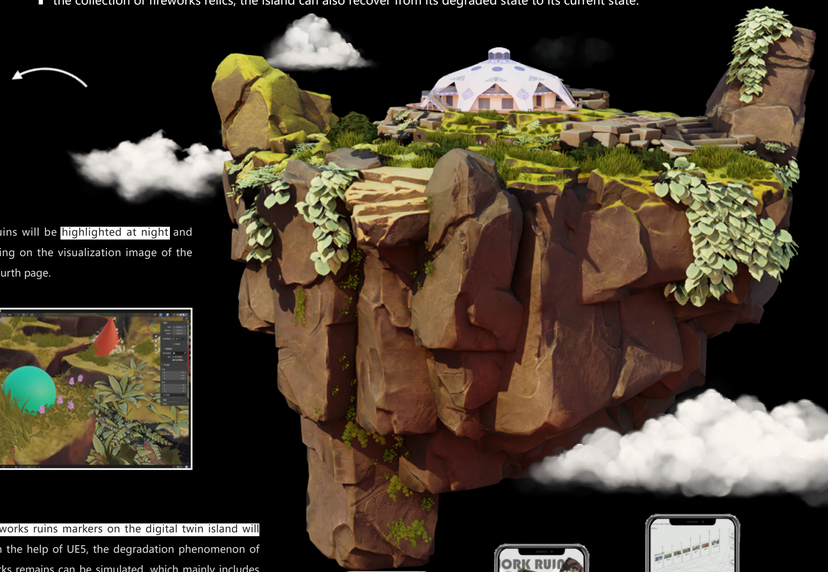


The top view below shows the distribution of different types of fireworks ruins on the digital floating island, which will change in real-time with the fireworks relics in the public welfare activity venue.



With the emergence of fireworks ruins, the vegetation of the digital twin island will also undergo corresponding changes, mainly characterized by yellowing and reduced vegetation.

The following picture shows the appearance of the digital twin island when it is not contaminated by fireworks relics. With the collection of fireworks relics, the island can also recover from its degraded state to its current state.



Uncollected fireworks ruins will be highlighted at night and can be viewed by clicking on the visualization image of the fireworks relics dyed on the fourth page.

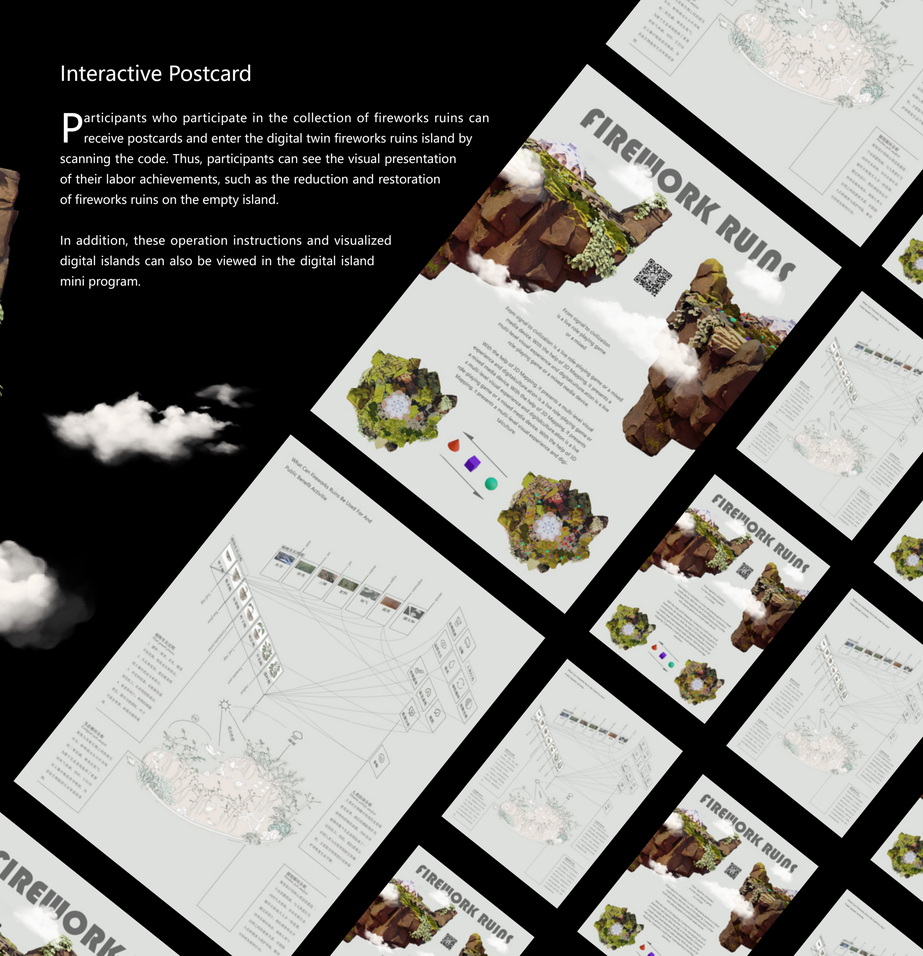
Vegetation degradation -

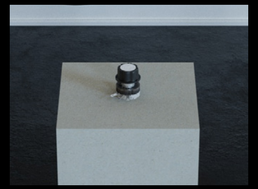
The vegetation near the fireworks ruins markers on the digital twin island will experience degradation. With the help of UE5, the degradation phenomenon of vegetation in video of fireworks remains can be simulated, which mainly includes three vegetation types: grass, dogstail, and shrub.

Interactive Postcard

Participants who participate in the collection of fireworks ruins can receive postcards and enter the digital twin fireworks ruins island by scanning the code. Thus, participants can see the visual presentation of their labor achievements, such as the reduction and restoration of fireworks ruins on the empty island.

In addition, these operation instructions and visualized digital islands can also be viewed in the digital island mini program.





VIRULENT BEINGS

BRIEF INTRODUCTION

The unemployment, poverty, and collective powerlessness brought about in resource depleted cities reflect the externalization of the "destructive impulse" in human subconscious. This project attempts to narrate through digital media, not only reproducing the material fate of these cities, but also revealing the deep psychological dynamics hidden behind human collective behavior, and guiding the audience to think about de futurism and the possibility of sustainability.

Cover And Background	P 01
Preliminary Investigation	P 02
Comparative Research And Video	P 03
Interaction Design	P 04
Video Storyboard	P 05
Work Display	P 06

Video link *https://youtu.be/6v3ZGif4eAA*

BACKGROUND

Inspiration

In the process of global industrialization, many cities that rely on a single energy or mineral often experience a typical trajectory of "rise fall transition". They rose due to resources and were once the core engine of national economic development.

However, with the excessive exploitation and consumption of resources, prosperity gradually declines, ecological environment is destroyed, social structure collapses, population outflow, industry shrinkage, and ultimately becomes a "resource depleted city". This process is not only an economic and environmental crisis, but also a psychological and cultural trauma.

Why Design

The current situation of de futurism

Unique research perspective

Collective powerlessness

People tend to overlook

Design Goal

Sustainable development concept

Explain the theory of life and death instinct

Visualising a deprived future

Immersive interactive experience

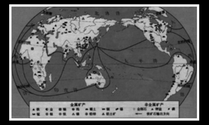
Metaphor Of Virulent Beings

The life forms referred to as virulent life are those whose very survival and reproductive expansion exert a destructive influence upon their hosts or the environment. The question we must consider is whether humanity itself falls into this category of virulent life. Does our pattern of survival and reproductive expansion also constitute a form of development that negates the future?

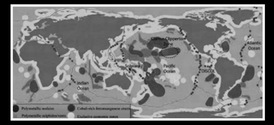
CURRENT SOCIATION

Life Cycle Of Resource - Based Cities

Resource - based cities often follow a typical trajectory of rise, decline and transformation. They rise to prominence through their resources, yet find themselves in difficulty precisely because of them.

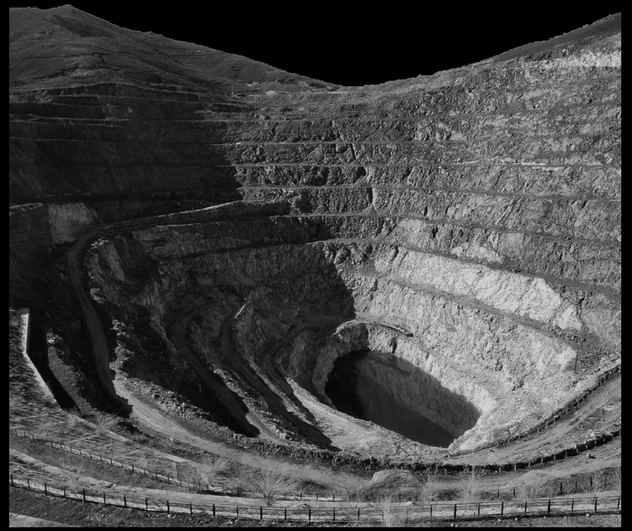


World's resource cities



More resource development

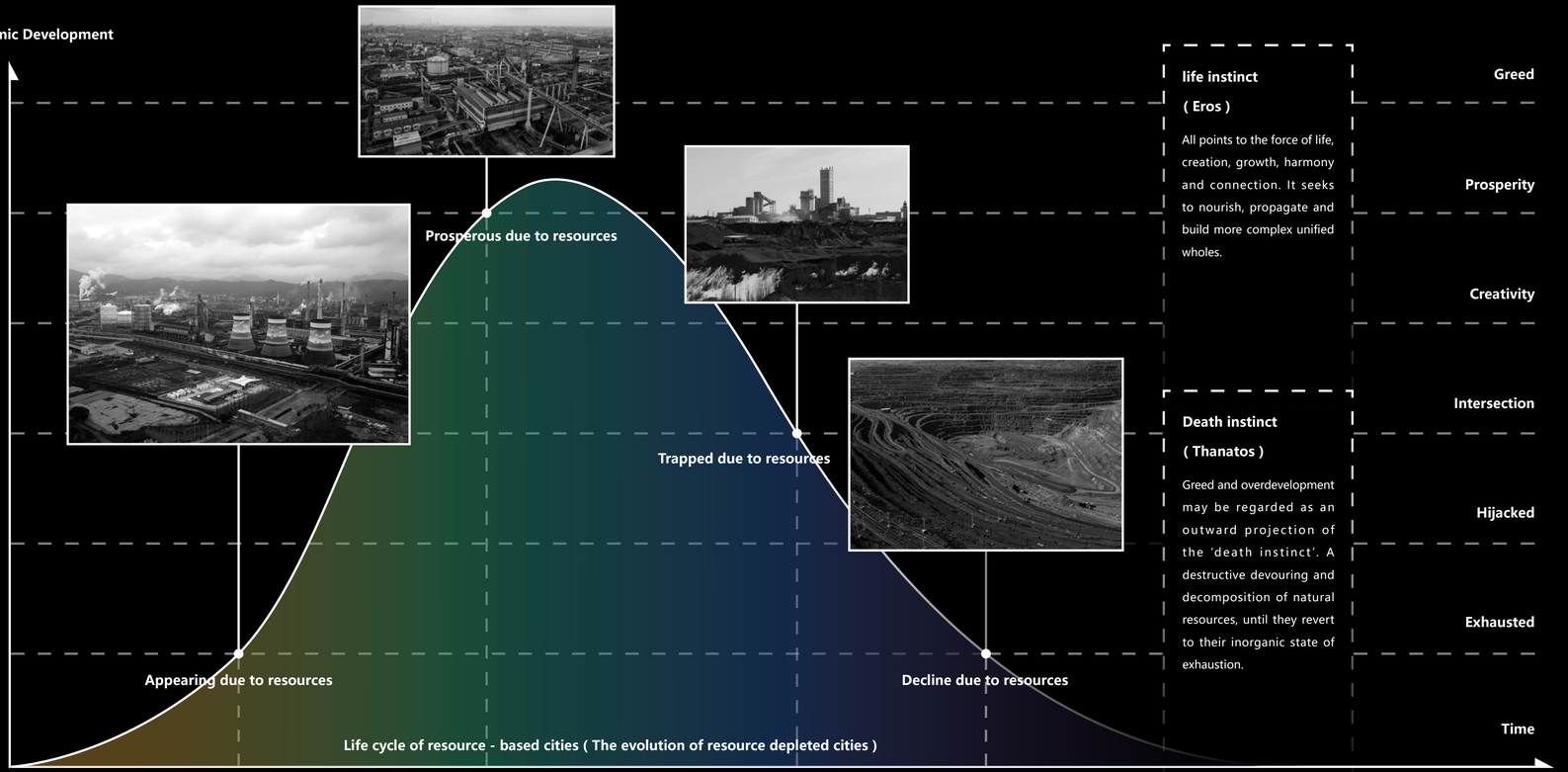
Destructive resource extraction does not stem from the malice of a few, but from an unconscious impulse that remains unchecked and unrestrained, pursuing only fleeting gratification and ruinous plunder.



DESK INVESTIGATION

The Evolution Of Resource Depleted Cities

The development of resource-depleted cities typically undergoes the following stages: *Appearing due to resources - Prosperous due to resources - Trapped due to resources - Decline due to resources.*



A Unique Perspective On Studying Resource Depleted Cities : Life Instinct And Death Instinct (Eros And Thanatos)

The theories of the 'life instinct' and the 'death instinct' were proposed by Freud. Viewed through this lens, the trajectory of resource-depleted cities can be interpreted as a contest between the life instinct and the death instinct. The tragedy of resource - depleted cities serves as a microcosm of how the "life instinct" within collective human behaviour transforms into a "death instinct".

life instinct

(Eros)

All points to the force of life, creation, growth, harmony and connection. It seeks to nourish, propagate and build more complex unified wholes.

Death instinct

(Thanatos)

Greed and overdevelopment may be regarded as an outward projection of the 'death instinct'. A destructive devouring and decomposition of natural resources, until they revert to their inorganic state of exhaustion.

Greed

Prosperity

Creativity

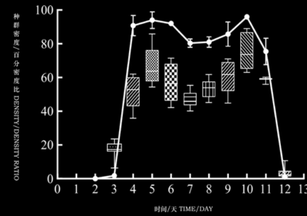
Intersection

Hijacked

Exhausted

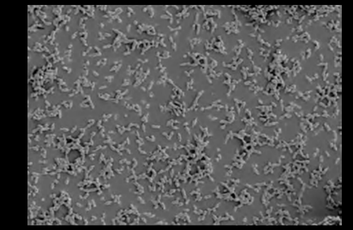
Instinct of virulent bacteriophage

If human fails to control its instincts and persists in pursuing strategies that undermine the future, its fate will mirror that of virulent bacteriophages.



The relationship between population density and time of virulent bacteriophages

Escherichia virus T4 under electron microscope -



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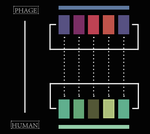
Comparative Research On Virulent Beings (Video Section)

Human activities

Rethinking the development model of the human industrial city implies a reflection on the critique of de-futurisation, which is the unsustainable unfolding of the city.



The T4 phage enters the cell by injecting its DNA and uses the host cell's mechanisms to replicate its own DNA and proteins, eventually causing the cell to rupture and releasing a new phage.

Enhancement and reproduction of virulent bacteriophages

Bentheim • Oilfield extraction

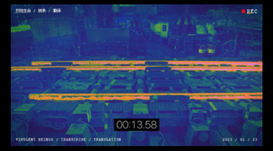
Resource acquisition (extraction of raw materials)

Human access to resources refers to the behaviour of human beings in acquiring various resources in their daily lives, and therefore the need for human beings to constantly seek, acquire and utilise these resources.

Adsorption

Adsorption is a prerequisite step for the phage to inject its DNA into the host cell by binding to host - specific proteins.

Zibo • Steel production



Resource transformation

Humans can transform raw materials into finished products, and the transformation of human resources is one of the most important signs of the development of modern society.

Transcription, translation (Biosynthesis)

Transcription is the flow of genetic information from DNA to RNA as the first step in protein biosynthesis. Translation as the second step in protein biosynthesis.

Shanghai • Automotive processing

Resource processing

More value and benefits can be obtained through processing. However, resource processing also faces some problems. Firstly, processing may produce pollutants and waste.

Assembly

The phage DNA, head and other components are synthesised separately, and finally the DNA shrinks and aggregates and is surrounded by the head proteins, followed by the gradual assembly of the tail.

Hohhot • Garbage releas

Resource development (searching for new resources)

Is humanity's pattern of survival, reproduction and expansion also a form of development that eschews future-oriented thinking, moving on to develop new resources once one cycle is complete?

Release

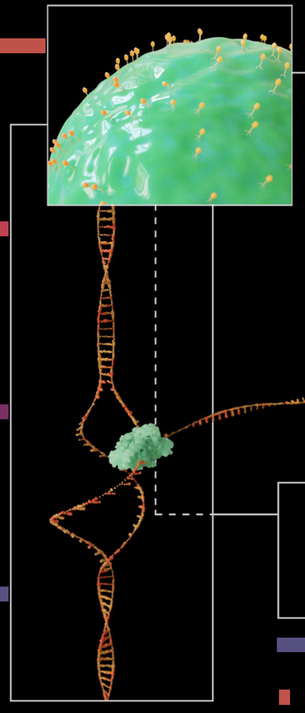
When the daughter phage reaches a certain number, the bacterial cell suddenly lyses due to the lysis of phage synthetic enzymes, and the released phage reinfects other sensitive bacteria.

Resource acquisition (extraction of raw materials)

Invasion

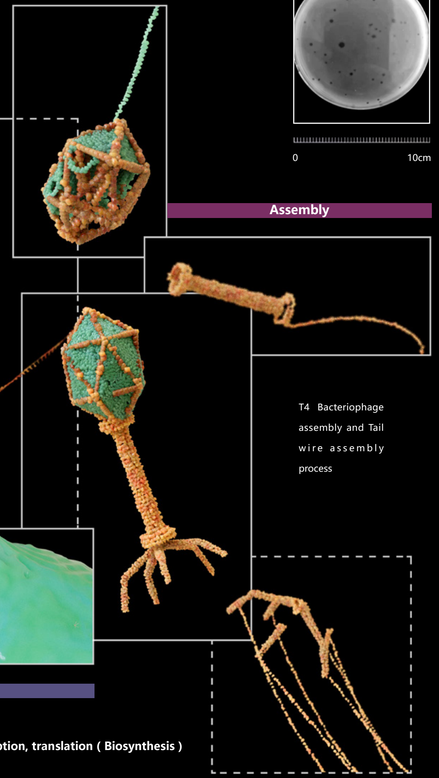
Virulent Phage Reproduction Sketch Map

Much like the predatory replication of phages upon their hosts, human society exhibits a similarly destructive impulse in the rise and fall of resource - based cities.



Biosynthesis

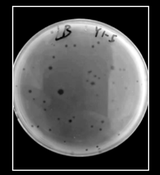
Assembly



Release

Adsorption
Transcription, translation (Biosynthesis)
Assembly
Release
Invasion

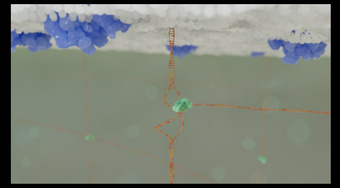
T4 phage culture dish



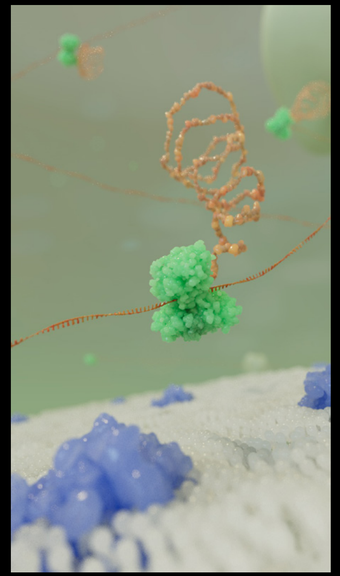
T4 Bacteriophage assembly and Tail wire assembly process

Assembly

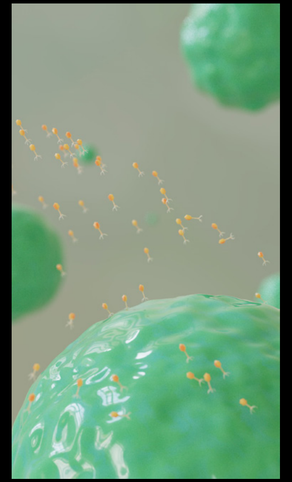
Invasion



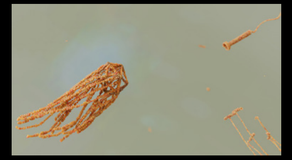
Transcription, translation (Biosynthesis)



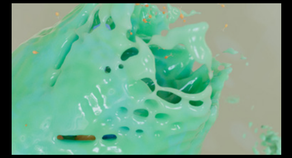
Adsorption



Assembly



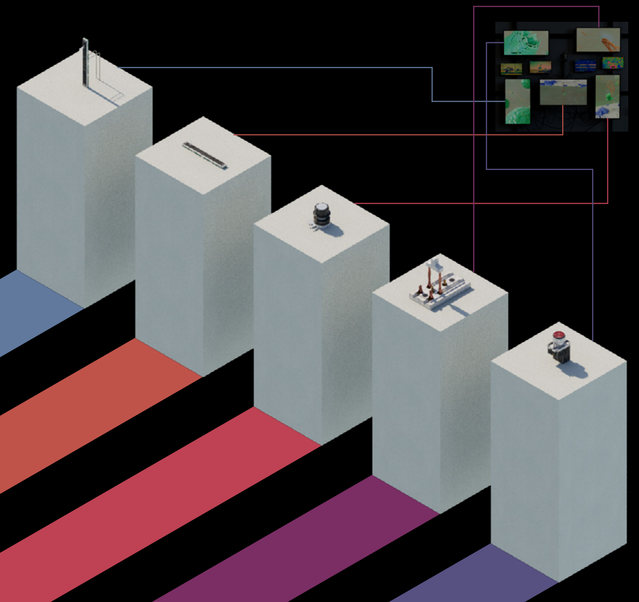
Release

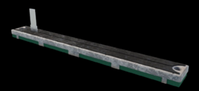


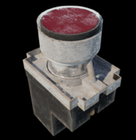
INTERACTION DESIGN

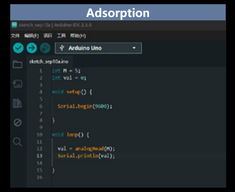
Interactive Device And 3d Experimental Video

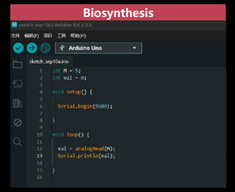
For the interactive elements of the visuals, five distinct actions have been selected to control the imagery, each corresponding to everyday and industrial components within the video content. These are: Injection (Invasion), Push (Adsorption), Rotate (Biosynthesis), Switch (Assembly), Button (Release).

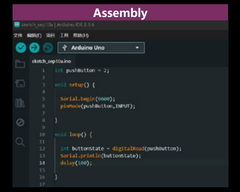


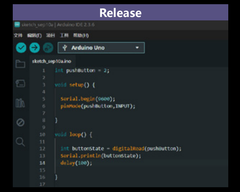












1、Injection - Pumping (Slider)

Simulate the injection process and oil extraction using pressing and pumping motions

2、Push - Pull (Slider)

Simulate variations in penetration or production intensity through a push-pull motion

3、Rotate (Turn up - turn down)

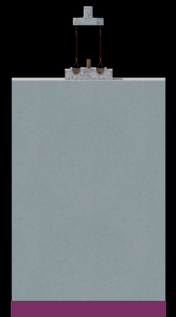
Rotary potentiometer action simulates adjustments in biosynthesis or production

4、On - Off (Switch)

The switching on and off of the switch simulates the commencement and interruption of the assembly process

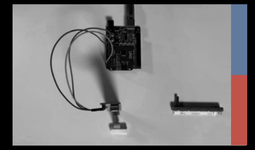
5、Press - Release (Button)

The button press action simulates scenarios such as sudden release and explosive tipping.

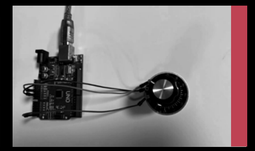


Injection - Pumping (Slider)

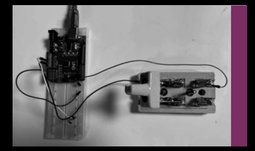
Push - Pull (Slider)



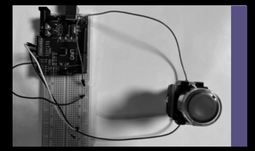
Rotate (potentiometer)



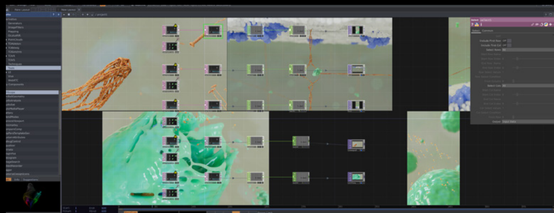
On - Off (Switch)



Press - Release (Button)



Arduino circuit connection and play control created with TouchDesigner -



STORY BOARD

01

F01 - F02 - A

Slider

Injection - Pumping

Push - Pull

F03 - B

Rotate

Turn up - turn down

Overload level

F04 - C

Switch

On - Off

By-product pollution

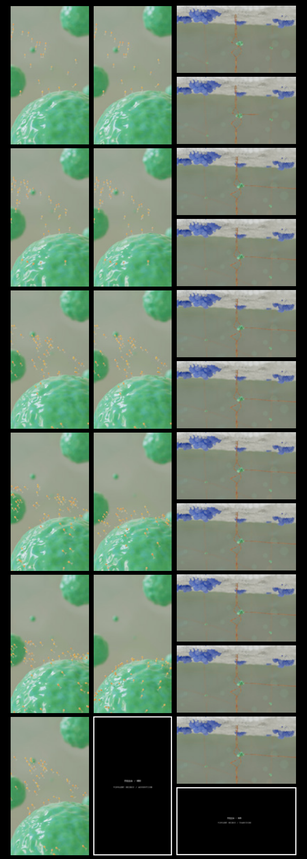
F05 - D

Button

Press - Release

Irresponsible emissions

02



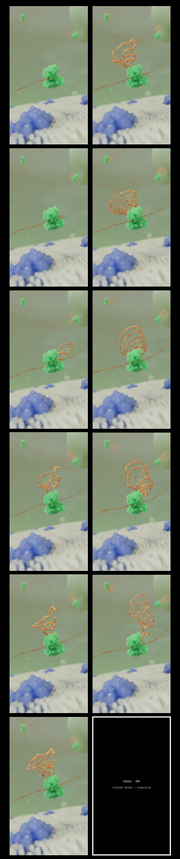
Witness humanity's relentless extraction of resources

Slider

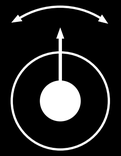
A

Oil extraction

03

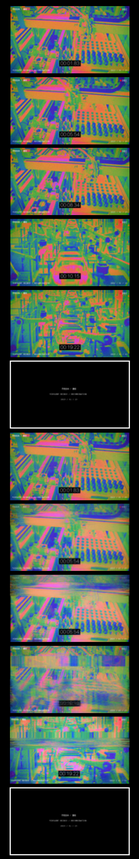


Experience the degree of overload in human production



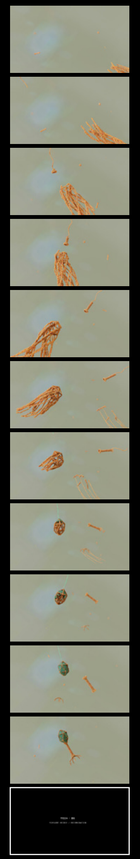
Rotate

B

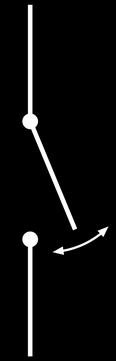


Disturbances in Resource Conversion

04

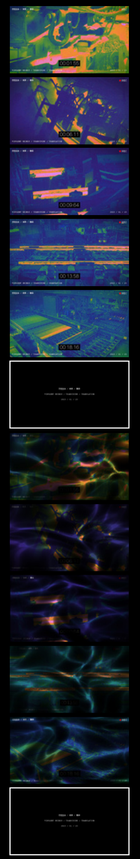


Experience the pollution caused by the by-products of human production



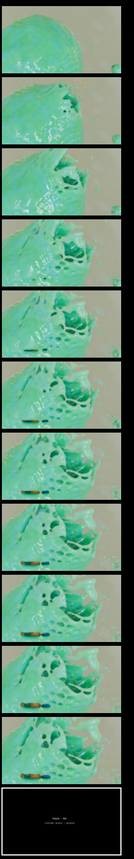
Switch

C



Waste generated during the manufacturing process

05



Experience humanity's irresponsible and emotionless emissions

Button

D

Discarded items

EXHIBITION

Display Details And Description

Interactive introduction

Five Types of
Large - Screen Interaction

Screen 1 - Adsorption

Interaction 1 - Injection - Pumping (Slider)

Screen 2 - Invasion

Interaction 2 - Push - Pull (Slider)

Screen 3 - Biosynthesis

Interaction 3 - Rotate (potentiometer)

Screen 4 - Assembly

Interaction 4 - On - Off (Switch)

Screen 5 - Release

Interaction 5 - Press - Release (Button)

Four Types of
Small - Screen Interaction

Screen A - Acquisition

Screen B - Conversion

Screen C - Processing

Screen D - Development

We can never know all the stories that go to the future, but we can construct the hidden or present connections in them. These 9 screens are used to present scenes of the processing of life, observing the unsustainable development and destruction, rebirth and annihilation in the microcosmic perspective of potent life.

The phage model at the heart of the apparatus, serving as a link between human industry and potent microscopic life.

Side view

Top view

Y 0 - 1 / Day

Rising period of population density index

Y 1 - 5 / Day

Stable period of population density

Y 5 - 8 / Day

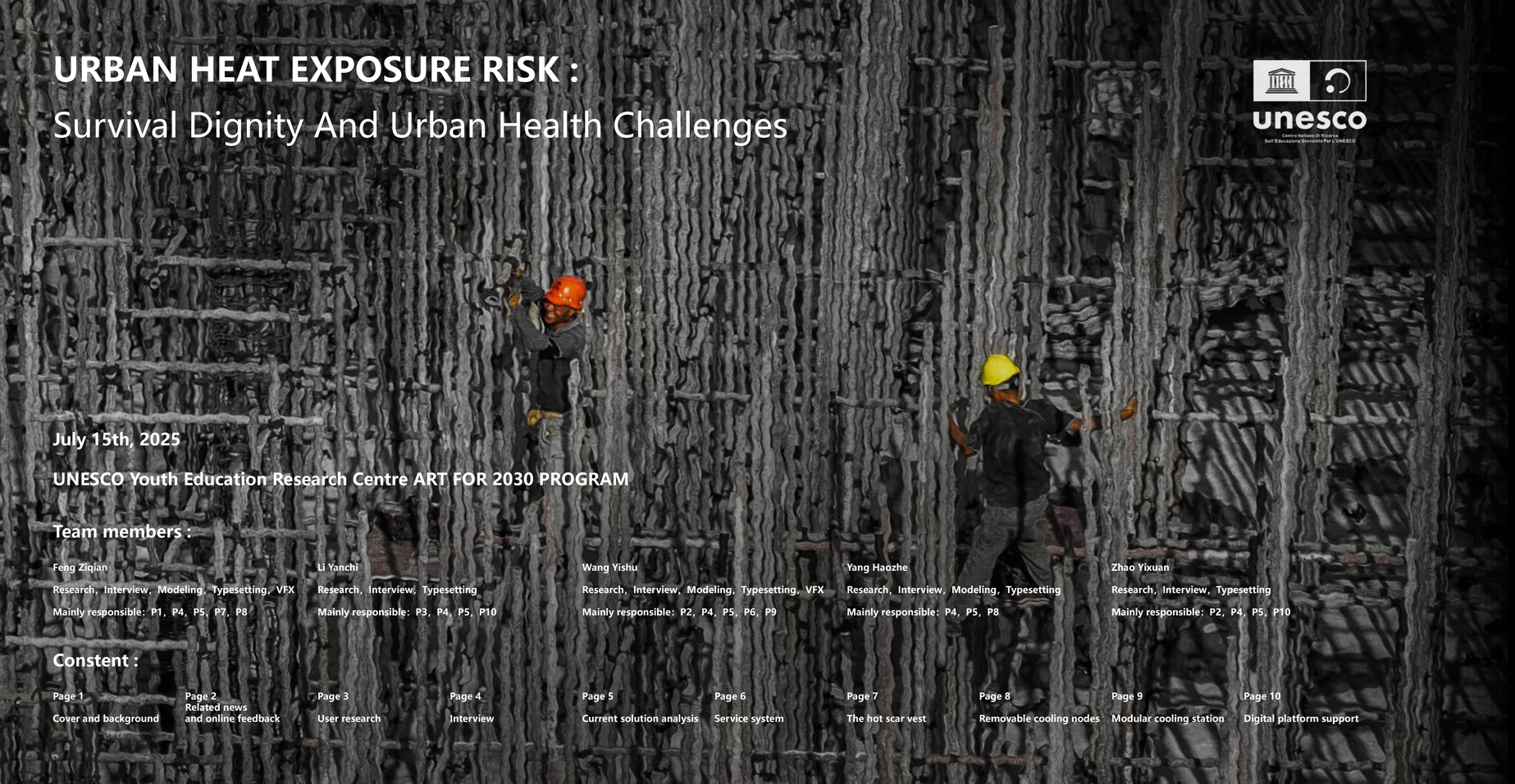
Rapid decline period of population density

Detail top view

URBAN HEAT EXPOSURE RISK :

Survival Dignity And Urban Health Challenges

unesco
Centro Italiano Di Ricerca
Sull'Educazione Giovanile Per L'UNESCO



July 15th, 2025

UNESCO Youth Education Research Centre ART FOR 2030 PROGRAM

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Constent :

Page 1 Cover and background
Page 2 Related news and online feedback
Page 3 User research
Page 4 Interview
Page 5 Current solution analysis
Page 6 Service system
Page 7 The hot scar vest
Page 8 Removable cooling nodes
Page 9 Modular cooling station
Page 10 Digital platform support

BACKGROUND

Topic And Inspiration

Urbanization process, coupled with climate change, global high temperatures, the urban heat island effect, and hardened surfaces, exacerbates the risk of heat exposure. Outdoor workers bear the brunt, facing various health challenges due to high-intensity physical labor, direct sunlight exposure, and limited protection

To alleviate the urban heat exposure crisis, it is urgent to focus on urban infrastructure and governance, improve high-temperature protective infrastructure, and propose systematic measures. Through inclusive technology and social collaboration, we aim to build a more inclusive and adaptable high-temperature protection system, safeguarding the vulnerable groups who are essential to urban operations and are exposed to heat.

Why Design	Design Goal
Life and health crisis	**Effectively cope with extreme weather**
Inequitable social structure	**Safeguard the legitimate rights of citizen**
Lack of system protection	**Provide effective protective equipment**
Vulnerabilities in urban systems	**Improve the urban heat-avoidance system**
Lack of social welfare	**Inclusive allocation of government resources**

Provide effective protective measures for high-temperature working populations through intelligent urban systems and technological products, and rebuild survival equity in a high-temperature society.

Definition of High - temperature work

High - temperature work refers to work under abnormal working conditions that combine high temperature, strong heat radiation, or high air humidity, with the wet bulb globe temperature index (WBGT index) exceeding the prescribed limit.

Workplace high-temperature operations refer to tasks where the average WBGT index at the workplace is ≥ 25℃ during the production and labor process.

Related News

How do people treat workers exposed to high temperatures?

01

Work at a temperature of 40 degrees for 4 hours

While traveling in Datong, a tourist dropped his watch into a trash can. He called the emergency number 12345 for help. Then, he asked two sanitation workers to search for the watch in the "garbage mountain" under the scorching 40-degree Celsius heat. It took them over four hours to find the watch. Various media also reported this incident. However, the hard work of the sanitation workers was not mentioned in the reports.

An elderly man died while working in a 40-degree heat environment

On July 6th, the temperature in the city exceeded 40℃. Three days ago, he still remained on duty for 11 hours each day. Before fainting, he had been exposed to the high-temperature working environment for a long time. The duty room where the dormitory supervisor worked lacking basic cooling equipment. When applying for a work injury recognition, he was also rejected by the school on the grounds that he was an "unofficial employee".

03

The worker died after working in the hot environment for 9 hours

When Wang Jianlu, an architect worker in Xi'an, unfortunately passed away due to heatstroke after working in a hot and humid environment for 9 hours, we didn't see any media voices expressing sympathy for his death. Instead, what we saw were posts like "#Hardworking workers under the scorching sun#" and "#Electricity workers eat watermelons at 79 meters high to cool off#".

Conclusion

These cases focus on incidents involving the public and high-temperature workers, reflecting the indifference of some members of the public towards the value and health risks of high-temperature workers, as well as the neglect by enterprises of providing adequate protection for employees, exposing them to danger.

Online Feedback

Why? Focus on the outdoor working scenarios under heat risks, capture the attitudes of society (social media / others) and the workers themselves towards heat risks, understand the cognitive differences from different perspectives, and the actual problems faced.
What do we understand? The working environment, potential risks and rights loopholes of outdoor workers under heat risks from the perspective of others; the workers' own understanding of high-temperature work, feedback on treatment, and the health and survival pressure caused by heat exposure.

Social Perspective

02

Empathy

The sweltering weather made it impossible for these migrant workers to breathe properly without the air conditioning on. After unloading the goods, they could only take a brief rest on the scorching roadside, looking sad.

Action

Today, I came to Ningbo to shoot the process of cutting reed grass. At 3:30 PM, the masters were working under the 38-degree heat. I filmed for a while and felt like I was about to explode from the heat. I rode to the village to buy 20 ice pops, and distributed them to the masters to cool off.

Empathy

Mom works every day in a stuffy work uniform. Today, after coming back, Mom said she felt dizzy. I said she must take off the work uniform because she felt unwell. Mom said it wasn't just her who was suffering like this. Others also endured it. I said that even when working, one can't just die from excessive heat. My mom, because the work uniform is very inconvenient to put on and take off. So she has requirements for going to the toilet. She doesn't allow going to the toilet and can't even drink water.

Care

The sanitation workers must be going through a tough time. On my way to work today, I saw this old man sitting on the roadside dozing off in the 30-degree heat. He had taken off his hot shoes, but there was not even a single shady spot for him to rest. It's truly a hardship in this world.

Worker's perspective

Lack Of Social Security

Recently, the platform has launched a heatwave care bonus program. If you complete the required sales volume within the designated period, you can receive a 20-yuan reward. However, it's quite difficult to achieve.

Lack Of The Courage To Speak Out

Our monthly salary is fixed. I haven't heard of any heat allowance. When the reporter introduced the heat allowance standard in Beijing to Grandpa Zhang, he said he was unaware of it and no one had ever told him about it. "Since no one has asked for it, I don't want to bring it up either."

In A Pickle

The workers are having a hard time. There's nothing we can do. Another young man said, "We thought we could just bear it for a while. A few days ago, a young man collapsed on the construction site. When he was sent to the hospital, there were no signs of life left."

No One Paid Attention

The workers shared stories of fainting due to the high temperature, being denied rest, and not getting enough water. "I want those who are important to know that this is our reality," commented a farm worker. "Our people are getting sick. We are thirsty. And it seems that no one cares."

Withstand High Temperatures

Today in Yantai, the weather was extremely hot, with the temperature soaring to 37 degrees. As a welder, working under such high temperatures was truly unbearable. To prevent being burned by the sparks, even though it was unbearably hot, I had to wear a thick long-sleeved work suit, a tight-fitting mask and gloves, and was completely covered up. During welding, the welding torch releases high heat, and combined with the 37-degree heatwave around me, I felt as if I were in a furnace. Sweat kept pouring out from my forehead and back, and soon my work clothes were soaked and stuck tightly to my body.

Conclusion

From a social perspective, attention is paid to the outdoor workers, who are "exposed to harsh working conditions in high temperatures + have weak rights protection". They are sympathized with and pitied.From the workers' perspective, they are forced to bear the heat risks due to their livelihoods. There are "blind spots in rights awareness" and "ignoring the dangers of heat" among them, and their health and survival are facing dual threats.

Questionnaire Survey for Public Portals

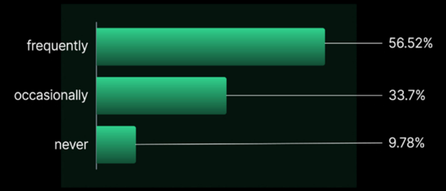
Questionnaire Design Logic Explanation

Cognitive Level (Questions 1-2):
Understand the public's level of concern and cognitive bias regarding the issue.

Action Level (Questions 3-7):
Quantify the public's actual willingness to help and their preferred forms of assistance.

Policy Level (Questions 8-9):
Explore the level of support for systemic solutions.

- Have you ever paid attention to the working conditions of outdoor workers (such as delivery personnel, sanitation workers, and construction workers) in hot weather?



56.67% occasionally pay attention—over half know outdoor workers' high-temperature conditions but not often. 33.33% frequently show stable concern. Only 10% never pay attention. Most are aware, with varying levels, indicating some social concern but needing more efforts to improve their work environment.

- What do you think are the greatest difficulties faced by workers in high-temperature environments? (Multiple answers)

- What is your acceptable average annual expenditure for assisting workers exposed to high temperatures?

Nearly a third of respondents would spend 50-200 yuan yearly to assist high-temperature workers— a moderate sum that eases financial strain while offering tangible help, reflecting public willingness and capacity. A small share would spend over 200 yuan, showing strong enthusiasm and support capacity. This provides an economic basis for related activities and suggests diverse donation options could attract more participants by fitting varying economic ability and willingness.

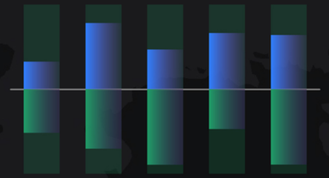
- During the hot summer months, have you ever actively helped outdoor workers (e.g., by giving them water or allowing them to delay deliveries)?

66.67% said "Yes," with over half actively helping outdoor workers in heat—like offering water or allowing delivery delays—showing majority concern. 33.33% said "No," for varied reasons. Some still need to boost support awareness, fostering broader societal care for these workers in extreme heat.

- If the community establishes a 'High Temperature Worker Aid Station' (providing drinking water and rest areas), would you be willing to participate in its maintenance or make donations?

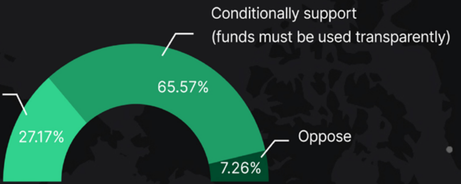
"Occasional supply donations" led at 48.89%, with nearly half willing to support aid stations—convenient, flexible, and low-burden. Regular participation (40%) showed many willing to invest time in maintenance, ensuring stable operations. Only 11.11% were unwilling, likely due to time or awareness issues. Most support aid stations, providing a strong public foundation.

- Which of the following ways do you think can support high-temperature workers, including both the support you yourself are willing to provide and the protections companies/platforms should offer?



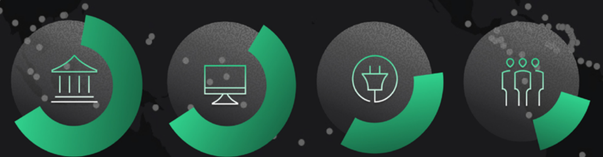
"Supporting government policies to boost high-temperature benefits" led at 78.89%, with 63.33% valuing supply donations and 47.78% willing to give tips, though service participation was low. Topping company expectations was ceasing work/rotating shifts (88.89%), plus high-temperature allowances, cooling gear (80% each), and rest stations (72.22%). Overall, government policy action and public contributions via supplies/tips, alongside corporate measures, guide improvements for these workers.

- Do you support improving conditions for workers exposed to high temperatures through taxation or public funds?



Most support using taxes or public funds to improve high-temperature workers' conditions but stress fund transparency and rationality, ensuring funds benefit workers, not for misuse. This reflects focus on proper public resource use and safeguarding workers' rights. Opposition is minimal, from concerns like fund efficiency or priorities. Data shows public support with a key focus on fund transparency.

- Which of the following measures do you think would be most effective in helping workers exposed to high temperatures?



"Government legislation mandating high-temperature protection" leads at 63.33%, with the public seeing legal safeguards as most effective. Technical solutions (15.56%) and corporate social responsibility (14.44%) are recognized by some, while public self-help (6.67%) is minor. Most view government legislation as key, so authorities should take the lead, combining approaches for comprehensive protection.

Conclusion

Overall, society has a certain degree of concern and willingness to support high-temperature workers, recognises the multiple difficulties they face, expects the government to enact legislation to protect them, and is willing to take action through material and policy support. In the future, support for high-temperature workers can be enhanced by strengthening institutional safeguards, broadening social participation, ensuring financial transparency, and innovating new technologies.

INTERVIEW

Interview For Workers Exposed To High Temperatures

The core purpose is to gain a comprehensive and in-depth understanding of the real situation of the high-temperature work population, people's cognition and attitude towards heat risk, actual effectiveness and demand gap of existing protective measures.

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Question 8	Question 9
Daily work schedule (e.g., whether to work during the high temperature period at noon),Average daily working hours, weekly working days ?	The high temperature characteristics of the workplace (outdoor exposure without shelter, indoor high-temperature equipment, etc.)	When working in high-temperature weather, the most common physical discomforts include dizziness, heatstroke, skin burns, dehydration, etc	Have there been any serious health issues caused by high temperatures? How about the subsequent recovery?	What chronic effects do you feel on your body after working in a high-temperature environment for a long time?	What high-temperature protective materials have been provided by the enterprise/unit? Are they sufficient?	Is there a mandatory rest system in place during work? (Such as halting work during high temperature periods) How effective is its implementation?	What additional protective measures would you take personally? What are the costs and effectiveness of these measures?	What do you think is the most lacking support for high-temperature protection at present?

Delivery drivers

Basic information
- Work in the city center
- Work experience: about 2 years
- Work for 12 hours per day
- Almost no days of rest

High-temperature issues
- Long duration of high temperature
- No protective materials provided
- Poor rest system
- The body is severely overstressed

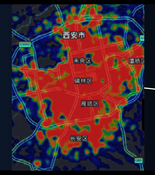
Needs and Expectations
- Dedicated resting place
- Upgrade protective materials
- A reasonable rest system
- Social respect and recognition

Cleaner

Basic information
- Work on the street
- Work experience: 10 years
- Work for 8 hours per day
- Almost no days of rest

High-temperature issues
- take every advantages of the cool
- Working is stuffy and hot
- Materials are inefficient and scarce
- Lack of hydration points

Needs and Expectations
- Sunshade, water replenishment
- Upgrade protective materials
- Be accepted by more venues
- Social respect and recognition

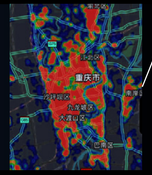


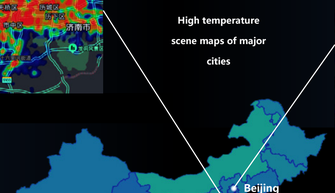
Gardener

Basic information
- Work at a school
- Work experience: 8 years
- Work for 9 hours per day
- Work for 6 days a week

High-temperature issues
- High frequency dizziness
- Inadequate personal protective
- Hurry to finish a task in time
- Live = Endure high temperatures

Needs and Expectations
- Upgrade protective materials
- Regular health check-up
- Implement high-temperature shutdown
- Social respect and recognition





High temperature scene maps of major cities

Basic information
- Work on the construction site
- Work experience: 20 years
- Work for 10 hours per day
- Optional rest days

High-temperature issues
- Live = Endure high temperatures
- Working is stuffy and hot
- Poor awareness of protection
- Meekly submit to oppression

Needs and Expectations
- Unable to improve
- Only need Higher subsidies
- Do not want to say
- Accustomed to high temperatures

Worker

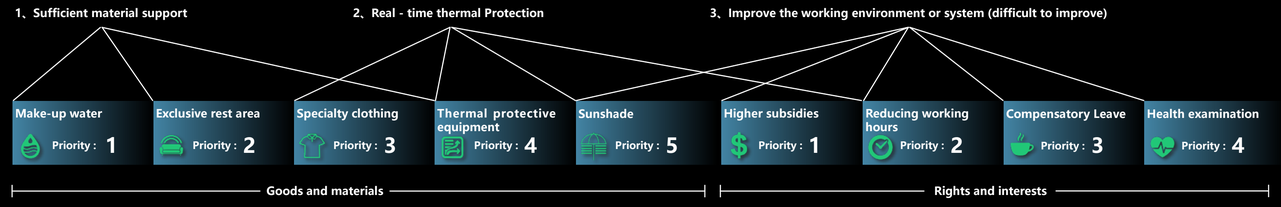
Basic information
- Work in a residential community
- Work experience: 4 years
- Work for 8 hours per day
- Work for 5 days a week

High-temperature issues
- No physical discomfort
- Community provides basic supplies
- Work even in high temperatures
- Meekly submit to oppression

Needs and Expectations
- Unable to improve
- Do not want to say
- It's not good to say it
- Can have Higher subsidies

security staff

The Main Relevant Methods To Mitigate Heat Exposure Through Interviews



Interview conclusions and summary

This interview analyzed the severe challenges and core needs faced by workers in high-temperature environments. The core issues lie in the inability to avoid high-temperature exposure, the lack of immediate and effective personal protective measures and emergency cooling methods, and the difficulty in changing the current situation (due to the nature of the work and site constraints).

The core solution approach focuses on personal protective gear, rapid emergency response, and solutions that ensure uninterrupted work.

1. Preventive basic protection (reducing calorie intake)
2. Proactive and rapid cooling system(core requirements: portable, instant, quick effect)
3. Emergency response support (public facilities and first aid)

CURRENT SOLUTION ANALYSIS

Interview For Workers Exposed To High Temperatures

Regarding the issue of urban heat exposure, many countries have taken measures and made various attempts at infrastructure, policy, and equipment technology levels.

1、Drone Thermography

The drone is equipped with an infrared thermal imaging device to scan the construction site during high temperature periods and detect abnormal body temperature of workers in real time. The drone reminds workers to rest through a speaker and automatically dispatches nearby unmanned vehicles to transport ice water and first aid kits.

Advantage: Real time performance and precise response
Disadvantage: Limited coverage

2、Ning Xiaofeng Station

Provide summer essential functions such as cooling off, dining and drinking water, emergency medical assistance, etc., and provide a "green shade" for outdoor workers in the surrounding area.

Advantage: Low cost of light asset operation mode
Disadvantage: Imperfect efficiency

3、Brumisateurs

It is a solar driven public spray cooling facility, mainly deployed in open spaces such as parks, squares, bus stations, etc.

Advantage: The cooling effect is significant
Disadvantage: Cannot be applied to workers

4、Cooling Singapore plan

The Singapore government has taken some measures to cool down the city, combining satellite remote sensing, ground sensors, and fluid dynamics models to guide long-term urban planning.

Advantage: Systemic heat island mitigation
Disadvantage: High cost and response lag

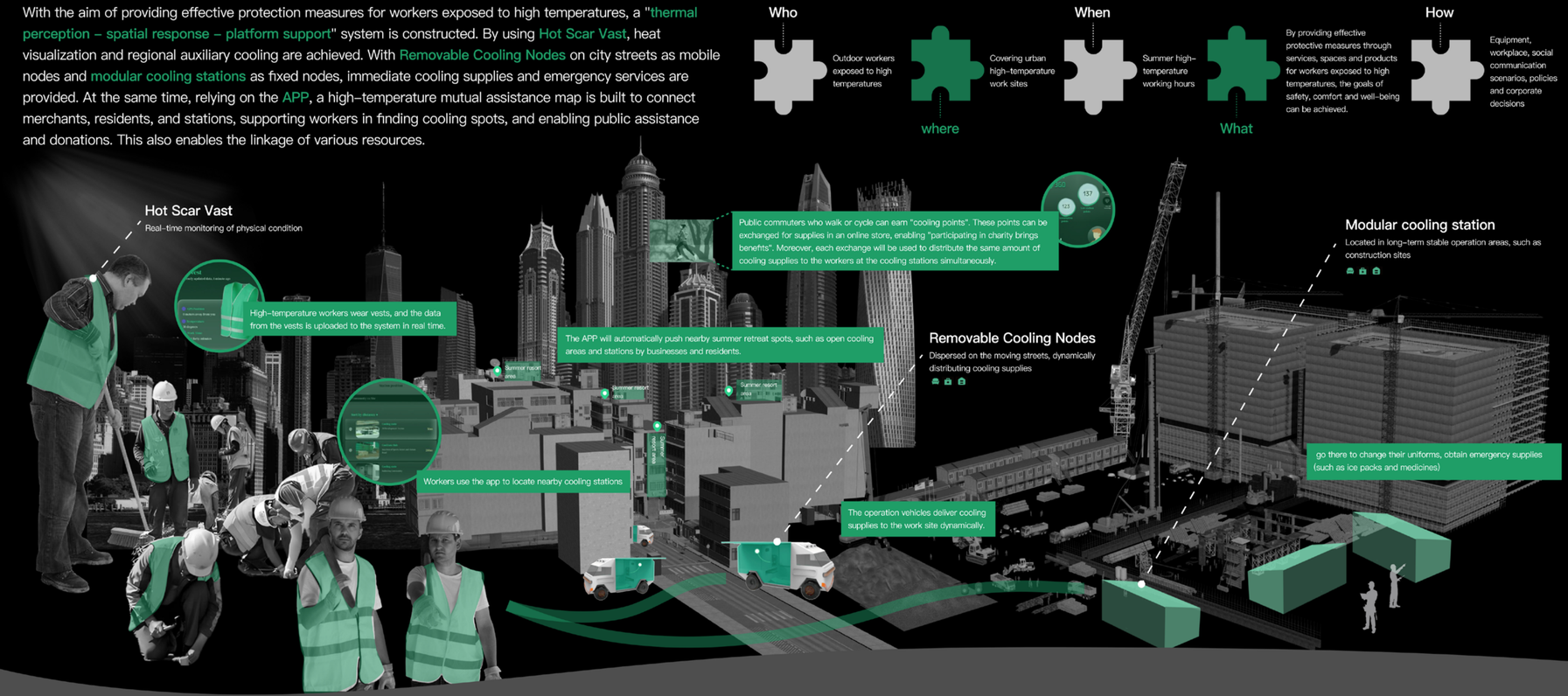
DESIGN ANALYSIS

Section	Discovery	Revelation
Related News	Clearly define that outdoor work groups are threatened by high temperatures, pose health risks, and have insufficient protection and rights protection; It also involves group characteristics, relevant data, as well as protective measures and existing problems.	We need to address issues such as high risk of heat exposure and difficulty in implementing protective measures. We need to focus on three directions: "proactive protection", "convenience and practicality", and "data empowerment rights".
Research at the societal level	**Outdoor workers are seriously threatened by high temperatures**	**Urgent need for convenient and practical proactive or mandatory protection**
Online Feedback	Social attitudes and feedback from various parties have shown that outdoor workers have poor high-temperature working environments and weak rights protection, but most of them are observed externally. Workers are forced to bear the risk of heatstroke due to livelihood reasons, resulting in a 'blind spot'.	1. Transforming the high temperature hazards of outdoor workers from "individual experiences" to "perceptible social issues" 2. Designing support systems that can seamlessly integrate into daily workflows 3. Low threshold, high participation social action connectors.
Research at the work environment	**Poor working environment for high-temperature workers**	**The Importance of Social Cognition, Daily Life, and Feasibility**
Questionnaire Survey	By using questionnaires to connect social perspectives, we can collect public perceptions, support perceptions, willingness to take action, and institutional expectations towards high-temperature workers, in order to design public participation paths and promote social collaboration in supporting high-temperature workers.	1、Building a public cognitive framework for high temperatures. 2、Enhance public empathy towards the target audience.
Individual research on high-temperature workers	**Analyze the core needs of high-temperature workers**	**Cooling down without affecting work, subsidies and more attention**
Interview For Workers	This interview analyzed the severe challenges and core needs faced by workers in high-temperature environments. The core issues lie in the inability to avoid high-temperature exposure, the lack of immediate and effective personal protective measures and emergency cooling methods.	For emergency, portable, fast, portable, flexible, immediate, uninterrupted work, and quick response to public facilities.
Design research on high-temperature workers	**Provide feasible systematic solutions or mitigation methods**	**Emergency, portable, fast, portable, flexible, immediate, uninterrupted work**
Current Solution Analysis	Advantages and disadvantages of existing solutions: Drone monitoring (Dubai): fast response but high cost; Spray pile (Paris): instant cooling but small coverage; Post station (Nanjing): stable but time-consuming, labeled; Urban heat island governance (Singapore): long-term but slow to achieve results.	1. Technical architecture design: does not affect the normal work of workers 2. Wide coverage 3. Centered on high-temperature workers: ensures dignity 4. Cost control design: time-sharing reuse.

Service System

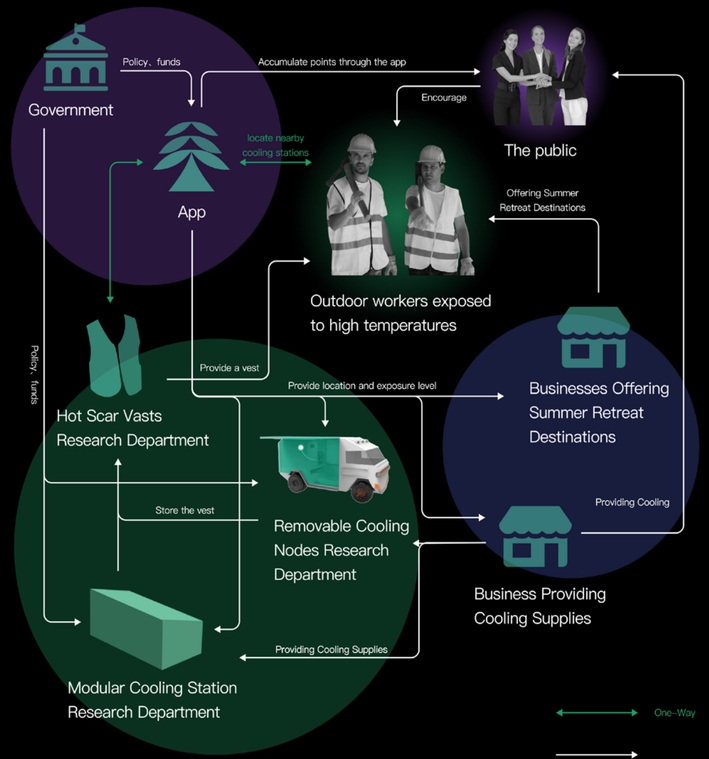
Interview For Workers Exposed To High Temperatures

With the aim of providing effective protection measures for workers exposed to high temperatures, a "thermal perception – spatial response – platform support" system is constructed. By using Hot Scar Vast, heat visualization and regional auxiliary cooling are achieved. With Removable Cooling Nodes on city streets as mobile nodes and modular cooling stations as fixed nodes, immediate cooling supplies and emergency services are provided. At the same time, relying on the APP, a high-temperature mutual assistance map is built to connect merchants, residents, and stations, supporting workers in finding cooling spots, and enabling public assistance and donations. This also enables the linkage of various resources.



Stakeholder Map

Interview For Workers Exposed To High Temperatures



THE HOT SCAR VEST

Thermal Visualization And Cooling Device

The Hot scar vest is an auxiliary system designed to provide real-time cooling for workers, aiming to alleviate hot working environments and enhance safety awareness by visualizing heat.

GPS positioner and main chip

Function ： Provide real-time positioning and display it on the map in the app for collection and emergency treatment.

Exploded view and functional description

Sensor mainboard and support frame

Function ： It provides temperature monitoring and vibration alarms, and transmits them to the app in real time.

Main body of vest

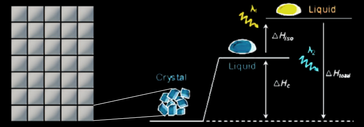
Function ： Cool down the user and display colors based on temperature in real time.

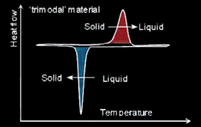
Elastic shoulder strap

Function ： The distance between the vest and the body can be adjusted for better cooling.

PCM (Phase Change Material)

PCM refers to the process where the state of a substance changes without a change in temperature, known as the phase change process. During this process, the phase change material will absorb or release a large amount of heat.





28℃ feels gentler on the body

Less prone to getting wet

Long duration

Outdoors 2 - 3 h / Indoor 3 - 4 h

High specific heat capacity

Accumulation of enthalpy change

12 - Color Temperature Variable Material Coating

Thermosensitive chromotropic paint is a functional coating material that changes color through different temperature changes.

Green

30℃

Suitable temperature

Suggestion ：

Green indicates that the vest is at normal operating temperature and the cooling effect is good.

Blue

32℃

Higher temperature

Suggestion ：

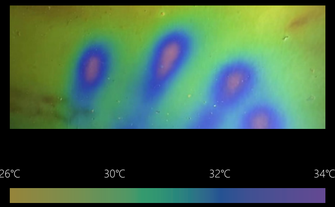
Blue indicates that the temperature is high or the vest has just become ineffective, requiring attention.

Purple

34℃

Failure temperature

Suggestion ： Purple indicates that the outdoor temperature is too high for work or the vest has become ineffective.



This color - changing substance is not only vibrantly colored but also capable of reflecting temperature changes in real time. If the internal cooling filler has lost its effectiveness, it will be indicated by a change in color.

Hot scar vest with heat marks comes in various colors, and there are corresponding suggestions.

Thermal Cycling Facility

Three types of thermal cycling devices are provided to match the Hot scar vest, adapting to different working environments, namely portable, vehicle-mounted, and stationary.

- **Portable and hand-carrying**
- **Insulation modemode for no power supply**
- **Cooling mode for power supply**
- **layered design**
- **Shared by multiple people**

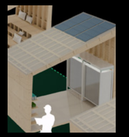
Wearing this vest is indeed much better than working in the heat before. Moreover, it's less cold and not particularly damp compared to the ice pack method I used before.

The replacement method of this vest is very convenient. I can simply take it from the portable refrigerator. Alternatively, I can find a nearby station to take a rest and then replace it.

These three thermal cycling systems can recharge the failed hot scar vest, making it usable again after a few hours. So a worker only needs to be equipped with two items.

In - car area

The car refrigerator can also provide replacement services for the Hot sacr vest, and can store supplies such as water and food.



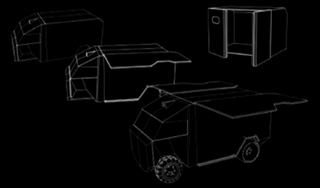
Modular cooling station

The Hot sacr vest change area is divided into storage and access spaces, facilitating workers to quickly find the appropriate jackets for replacement.

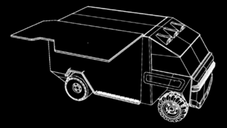
MOBILE SUMMER COOLING VEHICLE

Flexible Urban Summer Resort And High - Temperature Emergency Plan

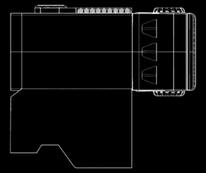
Mobile summer cooling vehicles are mainly prepared for high-temperature workers without fixed work locations. This urban networked escape and emergency system can efficiently serve the target population in the city with autonomous driving. This high efficiency and flexibility also provide wider usage space for hot scar vests. Moreover, its location can be displayed in real-time on the "High - Temperature Mutual Aid App".



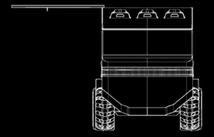
Sketch scheme



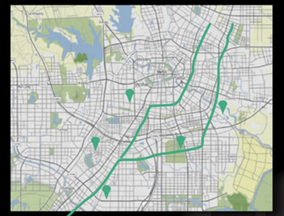
perspective view



top view



front view



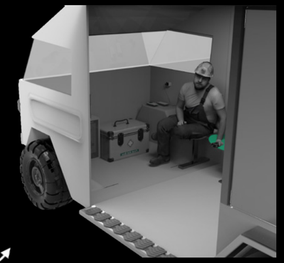
- Urban networked mobile summer vehicles provide workers with cool rest spaces
- Urban networked mobile summer vehicles provide emergency heat rescue for workers
- Urban networked mobile summer vehicles provide workers with replacement of hot scar vests

Urban system network

Unmanned trams that flow throughout the city make up the Removable cooling nodes. The location information of the hot worker who feels uncomfortable or suffering from heat stroke will be transmitted to the unmanned vehicle hub through the HOT SCAR VEST, and The H.E.A.T Sentinel will come to his side to rescue him. Routes are expected to appear in the visualization app.

Details Of Mobile Summer Cooling Vehicle

The vehicle adopts a modular design and is standardized with the equipment specifications in the entire service system, such as water, food, medicine, hot scar vests, and charging equipment. The interior of the vehicle is mainly divided into three functional areas: a large onboard refrigerator, an emergency stretcher area, and a rest area. The following is an introduction to the usage and detailed functions of different partitions.

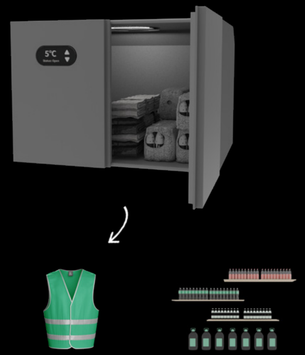


1. There is a row of seats in the car's rest area where workers can charge their phones

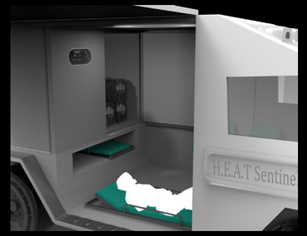
2. Workers can see the replacement suggestions of their own hot scar vest

3. Workers can see the outdoor temperature and summer work safety recommendations

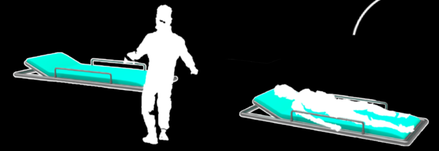
4. Workers can adjust the temperature inside the car to avoid discomfort



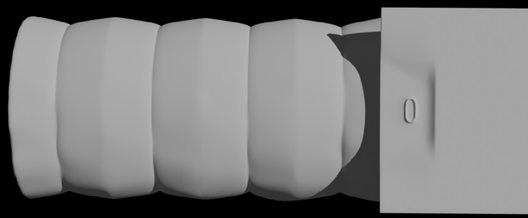
The most essential part of the vehicle is the large onboard refrigerator, which is equipped with replaceable hot scar vests, as well as water, beverages, and heatstroke prevention medication.



The car is equipped with emergency medicine and can accommodate one to two workers with mild heatstroke lying flat on stretchers.



Vehicle mounted stretchers are prepared for individuals with mild and severe heatstroke. Mild heatstroke workers can rest and take medication inside it, while severe heatstroke patients can wait for an ambulance inside the vehicle.



retractable sunshade cabin

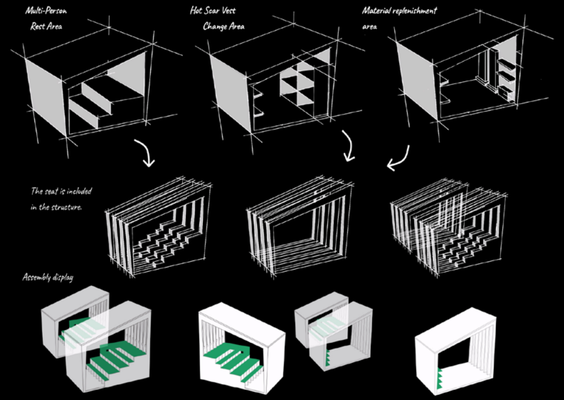
The retractable sunshade cabin can be installed in specific situations to expand the cooling space of cooling vehicles and accommodate more workers.

MODULAR COOLING STATION

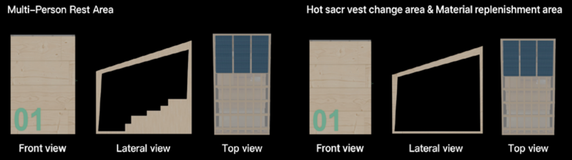
Brief Introduction :

In response to the demands of dynamic site layout, rapid disassembly and assembly, it takes sheet modularization and quick assembly as the core. It is easy to assemble and disassemble, and can be stored and transported after disassembly, with reusable residual value.

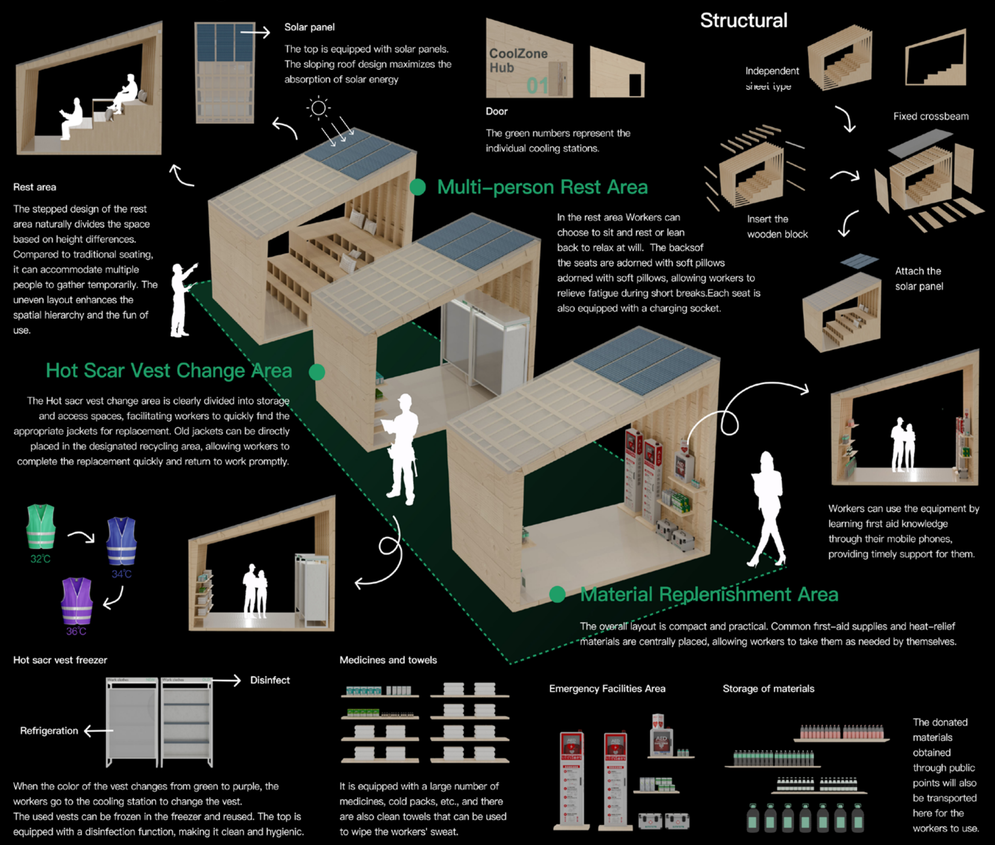
Draft



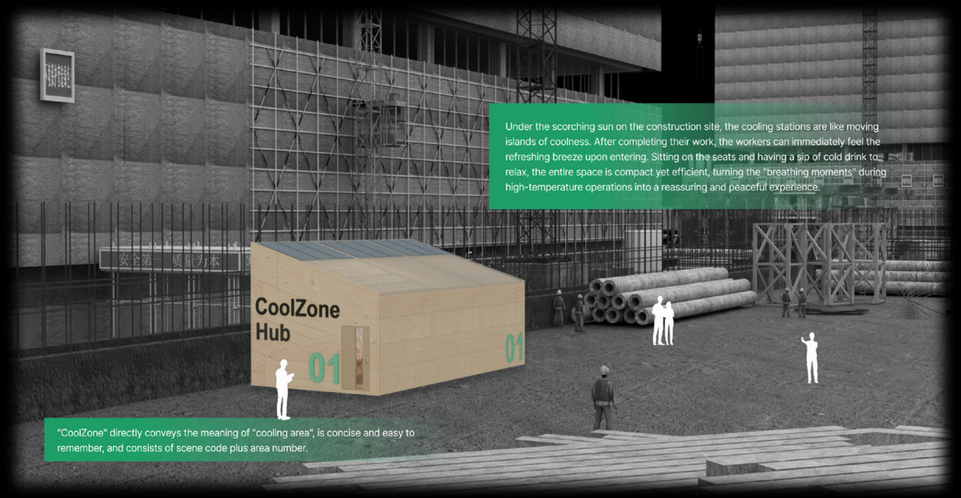
Three view drawing

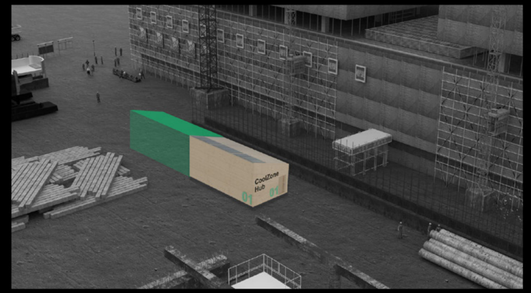


Detailed Design



Usage Scenario





The Cooling Stations Are Located At The Periphery Of Each Operation Area Of The Construction Site. Whether It's For A Short Break, Emergency Handling, Or Material Replenishment, They Can Be Efficiently Adapted To, Becoming "Cooling Guarantee Points" For The Construction Site During Hot Weather Operations.

DIGITAL PLATFORM SUPPORT

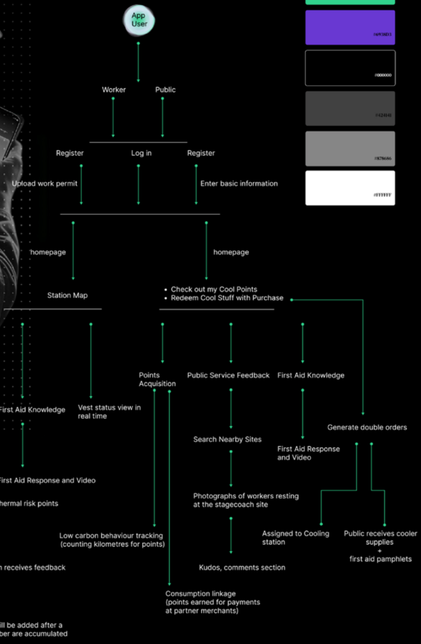
High - Temperature Mutual Aid App

Brief Introduction :

Developing a high - temperature mutual - assistance mini - program can help the thermal - mark vests be used more efficiently, improve the layout of the mobile space station, increase public participation, form a public - welfare closed - loop, optimize projects with data, and promote the development of high - temperature protection work.

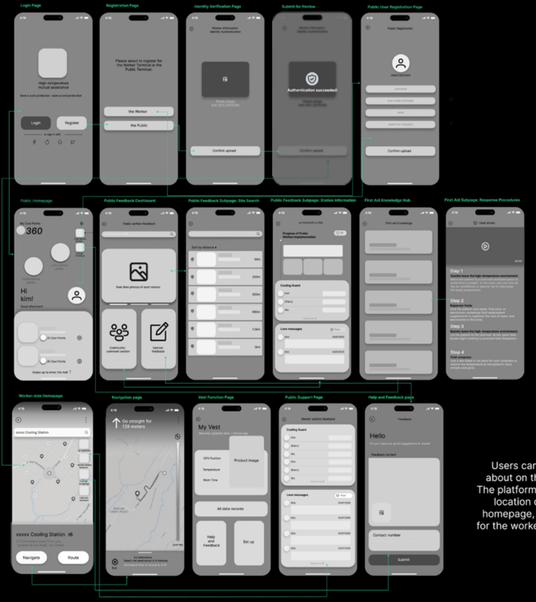
Font and Color Specification

Feature Structure



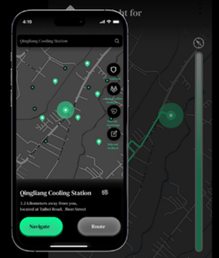
Low - fidelity prototype diagram display

The three sections belong to the login/registration section, the public end, and the worker end respectively:
Login/Registration Page: With a simple form, it enables quick account creation/login, paving the way for users to use functions.
Public End Page: It has a simple layout, aggregates services such as cooling stations, etc., facilitating citizens to quickly search and use.
Worker End Page: It focuses on task and status functions, helping with efficient operation.
The three cover different users, match needs with clear functions, and jointly build a high-temperature assistance service system.



Users can search for the summer retreats they want to learn about on the search page and view their charitable feedback. The platform will also recommend nearby retreats based on the location data you have provided. By clicking on the retreat's homepage, you can leave words of praise and encouragement for the workers resting there, allowing kindness to be conveyed through the coolness.

Worker Side Homepage



The worker app focuses primarily on the map navigation system. The map marks nearby cooling stations and cooling-friendly shops, making it easy for workers exposed to high temperatures to navigate to them.

This page helps workers view information related to smart vests, including their GPS location, real-time temperature, working hours, and complete data history.

Public Side Homepage

MY COOL POINTS

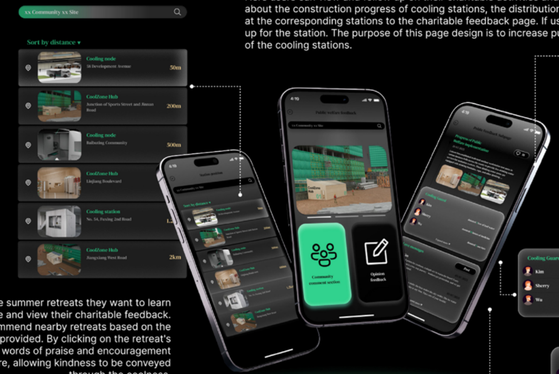
The first section allows users to see their cool points related situation. It clearly shows the total number of Cool Points, and the two branches of Low Carbon Points and Consumption Points. This design allows users to have an intuitive understanding of the source and quantity of their points, which is convenient for redeeming in the Points Mall.

ONLINE POINTS MALL

At the bottom of the page is the supplies exchange area. It shows some cool materials that can be exchanged with points in the online mall, and you can enter the mall page by sliding up the bottom of the page. Each time a user places an order, the platform will automatically generate a double order, one will be accompanied by a booklet on first aid knowledge in the face of high temperatures, and the other will be donated to the corresponding cool station in the user's name.

Public Feedback

Here users can view and follow up on their charitable activities and feedback. The platform will gradually upload information about the construction progress of cooling stations, the distribution of cooling supplies, and the rest conditions of workers at the corresponding stations to the charitable feedback page. If users are satisfied with this, they can also leave a thumbs-up for the station. The purpose of this page design is to increase public participation and better monitor the implementation of the cooling stations.



The details page of each site will display a scrolling list of user avatars and names who have donated cooling supplies to that site, as a token of our gratitude. This also makes our online cooling charity campaign more transparent.

Users can also leave heartwarming messages for workers exposed to high temperatures on the website. All messages will be synchronised to the workers' app, ensuring that kindness is conveyed not only through material support.

PROPHECY MANTRA

TIME PLANNING PRESCRIPTION AGAINST INSTRUMENTAL RATIONALITY

NEW MYTH NARRATIVE DISPELS DIGITAL SELF TRACKING AND ANCHORING VALUE RATIONALITY IN THE DATA FLOOD

BRIEF INTRODUCTION

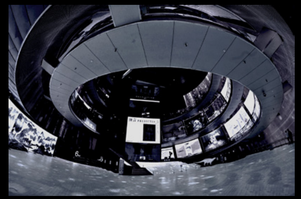
Brief Introduction

Nowadays, digital devices and platforms have sparked a widespread phenomenon of self tracking. This work aims to use EEG signals to provide predictive time planning suggestions and use new mythologizm to oppose the extreme expansion of instrumental rationality and squeeze the space of value rationality. Dealing with and alleviating the helplessness of pessimistic emotions and self sabotage.

Exhibition Venue:

China National Film Museum （Beijing）

The work was exhibited at the China Film Museum in July 2024 under the theme “Breaking Free.” It pays homage to the film Arrival while also honoring Sapir-Whorf hypothesis——the theory linking language to thought patterns, thereby offering participants new ways of thinking and acting.



Why Design

Instrumental rational expansion

Self tracking and quantitative discipline

Subject deficiency in human-machine relations

Emotional anxiety in technological dependency

Design Goal

Thinking and expanding value rationality

Safeguard the legitimate rights of citizen

Vent pessimism in work and life

Alleviate self - inflicted mental exhaustion

Instrumental Rationality Surpasses Value Rationality

Value Rationality < **Instrumental Rationality**

Emphasising the guiding role of moral judgement, personal beliefs or aspirations in shaping behaviour.

Typically associated with goal - oriented behaviour, placing greater emphasis on outcomes rather than values.

Self Tracking And Quantified Self

Quantified Self was first coined by Wired journalists Gary Wolfe and Kevin Kelly to describe the practice of using digital devices to quantify, track and manage one's own health and well-being. Those who engage in this behaviour are termed “self-trackers”. Quantified self, as a fusion of modern technology and lifestyle, provides tools for self-management and optimisation while presenting individuals with new challenges and issues.

DESK INVESTIGATION

Current Sociation

Global health app users have reached 4.5 billion, with smartwatch usage surpassing 100 million. 'Sharing step counts,' 'sleep scores,' and 'calorie tracking' are emerging as new forms of social currency. This phenomenon represents a digital evolution of Foucault's biopolitics.

The Alienation Carnival Of Data Body

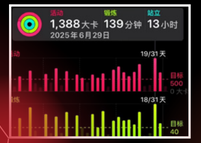
When human metabolic data becomes the core asset of commercial platforms, our biometric traits undergo a "double alienation". Training logs transform muscle soreness into virtual badges of honour. Individuals seeking freedom willingly don more sophisticated shackles of data.

Digital Self - Governance Techniques

Disciplinary power has shifted from physical spaces to algorithmic ones. The achievement badge systems in fitness apps are essentially digital versions of disciplinary reward mechanisms, rendering users as addicted to data progression as gamers.

New Mythological Redemption

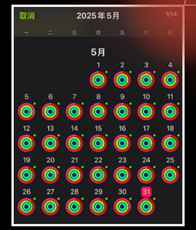
Quantified self serves both as a tool for liberating our cognition and as a novel cage of alienation. Rather than using quantified self to combat existential anxiety, we would do better to temporarily cast off the shackles of data discipline.

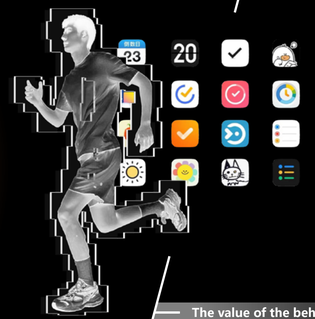


SELF TRACKING

GOALS

OUTCOME





The value brought by behavior

If it rains outside and I don't go for a run, I'll miss that medal.

- The essence of instrumental rationality lies in mastering calculation: determining costs and benefits against specific objectives to identify optimal means, while disregarding personal emotions and preferences.

The value of the behavior itself

If it's raining outside, I can do some other exercises at home.

Value - rational judgements possess no definitive answers, containing greater subjective elements. Individuals may focus more on the act itself, potentially diverging from the collective societal value judgements. The link between medals and health is not as strong.

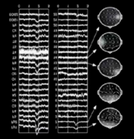
EMOTIONS AND WISHES

DESIGN PROPOSAL AND ANALYSIS

Visualisation Of EEG

EEG signals carry an individual's rich emotional landscape. Through emotional connection, we may glimpse more direct, even illogical psychological patterns. By interpreting the generated charts ourselves, we ultimately achieve self-guided resistance against disciplinary control.

EEG signal measurement



EEG signal analysis

Weight distribution

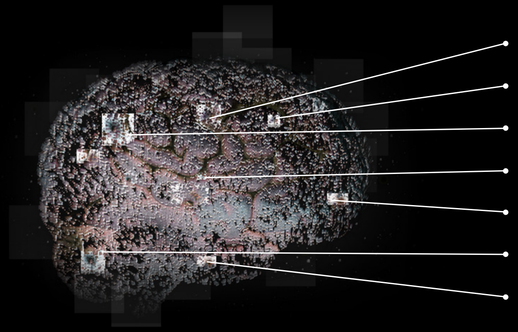
Scheduled time

Visualization

Timetable

Real - time changes

Electromagnetic control of ferromagnetic fluid



High Gamma (32 - 100 Hz)

Low Gamma (32 - 100 Hz)

High Beta (13 - 32 Hz)

Low Beta (13 - 32 Hz)

High Alpha (8 - 13 Hz)

Low Alpha (8 - 13 Hz)

Theta (4 - 8 Hz)

Calm and tense (+50%)

Excitement and Fear (+25%)

Pressure and focus (-25%)

Depression and meditation (-50%)

6: 00 —— 8: 00

8: 00 —— 10: 00

10: 00 —— 12: 00

12: 00 —— 14: 00

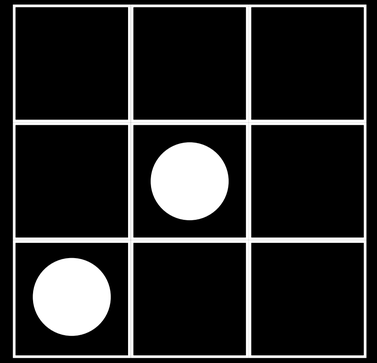
14: 00 —— 16: 00

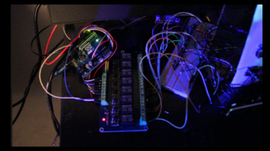
16: 00 —— 18: 00

18: 00 —— 20: 00

20: 00 —— 22: 00

22: 00 —— 24: 00



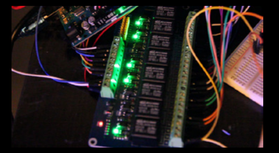


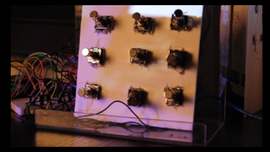
EEG signal analysis and the circuit

Analysis of brainwave signals primarily relies on an electroencephalograph (EEG) to transmit data to TouchDesigner for subsequent processing.

Relay switch control

Use a relay switch to control whether the electromagnet is energized and operational. This allows the Arduino's low current to control the high-current electromagnet.



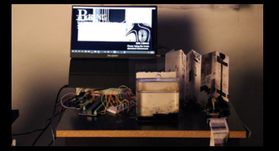


Electromagnetic control of ferromagnetic fluid

Nine electromagnets control the state of the ferrofluid. Due to the ferrofluid's unique properties, each flow produces distinct visual effects.

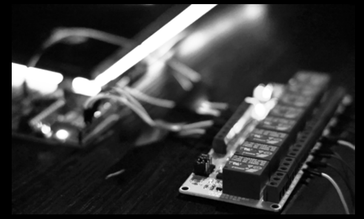
Visualization and printing

When the participant presses the print confirmation button, the thermal printer will print the current status image and text content via the Arduino.



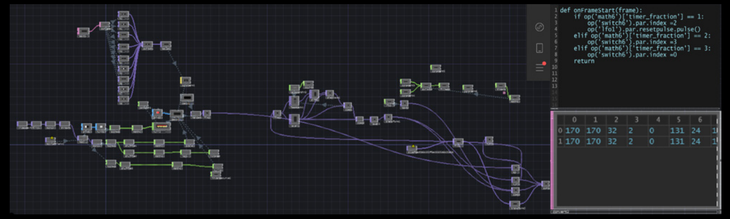
Measurement and analysis of EEG signals

Direct measurement of brainwave signals seeks to harness subjective emotional judgements about a given matter to filter out disciplinary noise introduced by external applications or platforms, thereby curbing the proliferation of imposed instrumental rationality.



Simulate the 'Transformer' algorithm

Different brainwave signals correspond to distinct emotional states, which are assigned meaning to each grid cell in the form of weightings. Moreover, the emotions evoked by observing the grid cells continue to influence the brainwave signals.

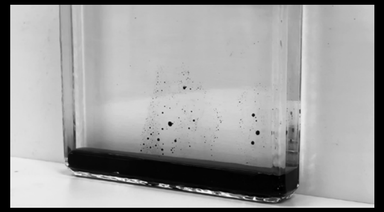


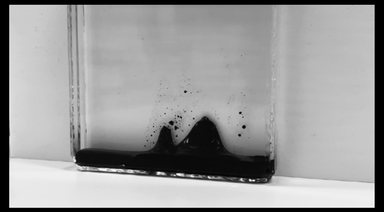
DESIGN PRESENTATION

Physical Objects And Participant Experience

The transformations of the ferrofluid mirror the shifting thoughts and emotions of the participants. They may continue to experience this mythical, ritualistic flow and metamorphosis, or press the print button to capture this fleeting moment.

The printed timetable may serve as a suggested schedule for a particular undertaking, aiding in determining when to undertake it or whether to proceed at all based on one's subjective feelings towards the matter.





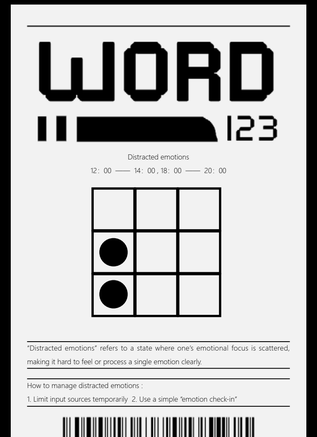
Participant feedback

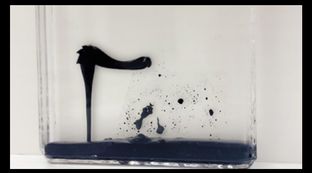
The prophecy itself may not be accurate, but these ritualized experiences have given me new insight into what I must do —— and they serve as a way to alleviate my anxiety.....

Design manual

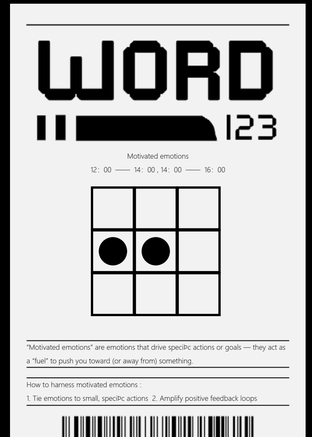
The design manual is available for visitors to browse during the exhibition, providing deeper insights into the works and guiding how to interact with or interpret the prophecies.

Distracted emotions **01**

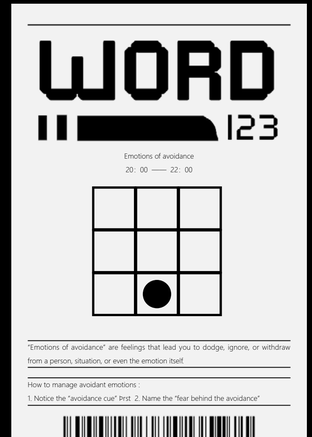


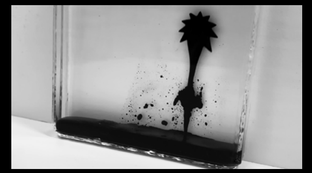


Motivated emotions **02**

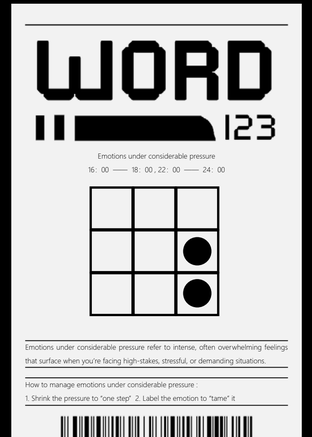


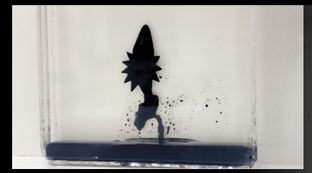
Emotions of avoidance **03**



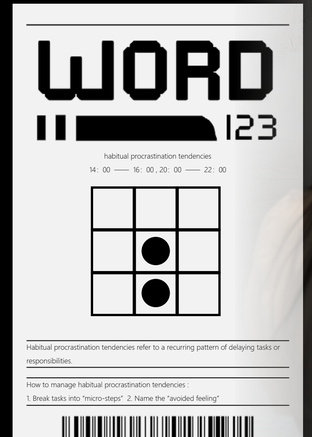


Emotions under considerable pressure **04**





Habitual procrastination tendencies **05**

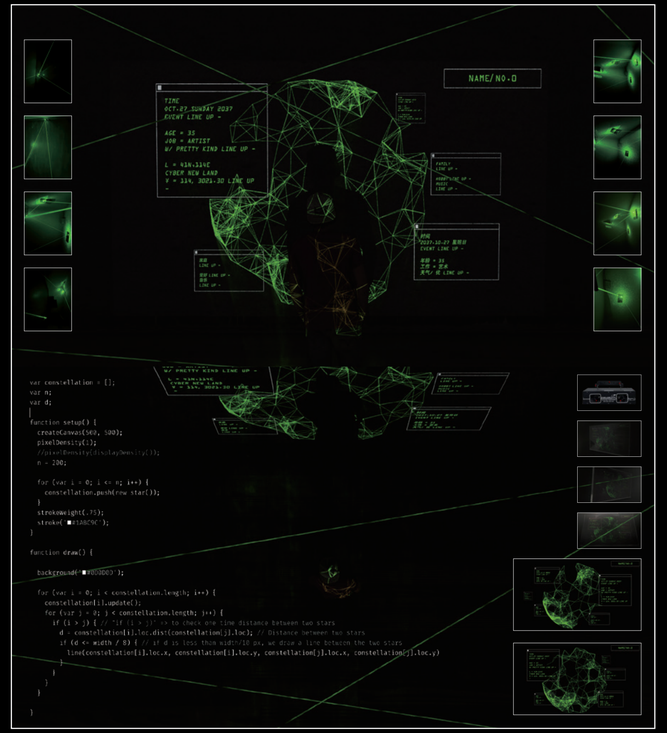


THE REALM OF HYBRIDITY

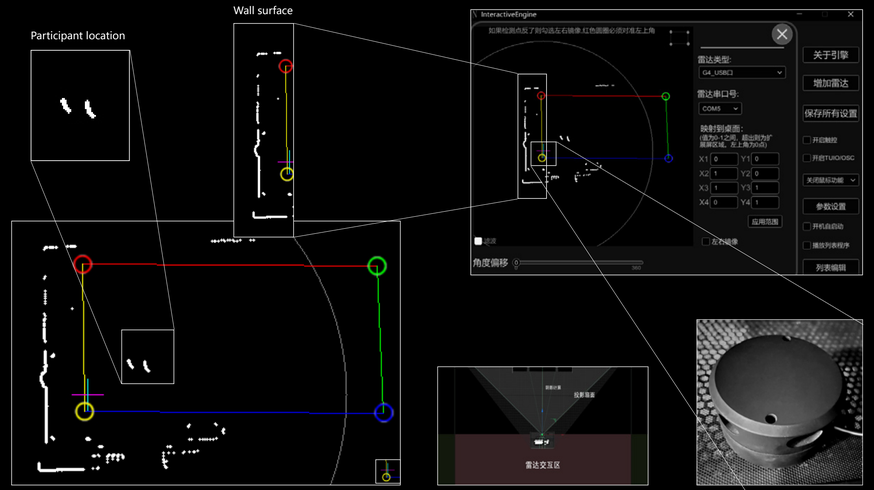
Brief Introduction

The Realm of Hybridity is a digital world driven by an "emergent" algorithm, where participants may alter two variables —— "separation" and "alignment" —— within pre-set scenarios by providing positional coordinates via LiDAR. Thus, simple rules construct an interactive, abstracted digital world, enabling audiences to experience and contemplate complex systems and the spontaneous emergence occurring around us.

Other Works **06**



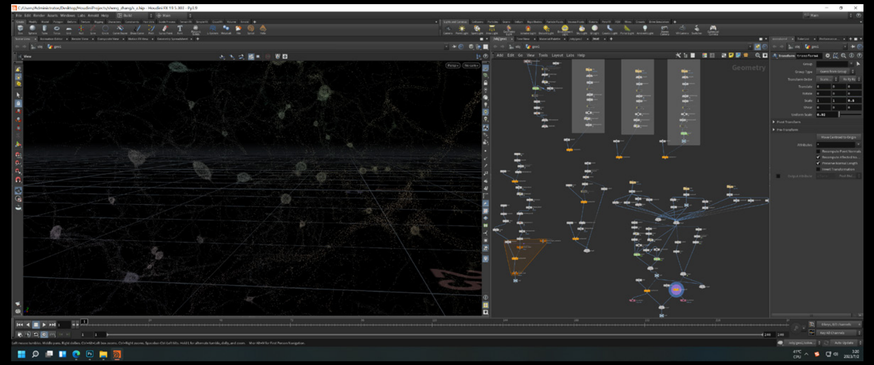
Interaction Controlled by Lidar

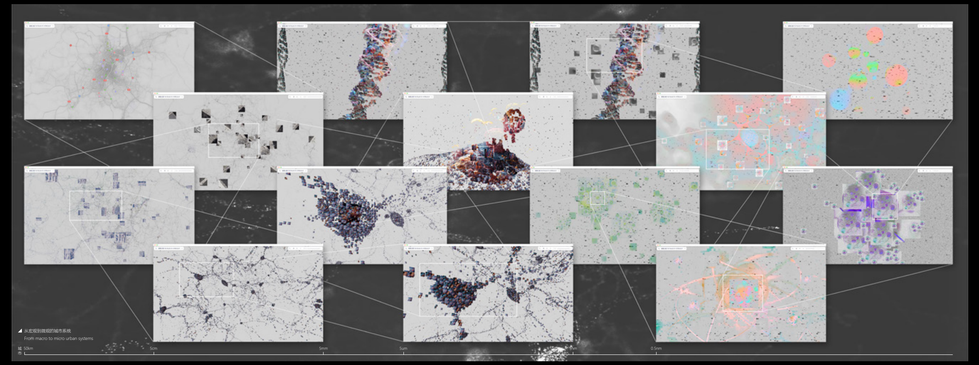


LiDAR coordinate map（ radius = 14 - 16 m ）

Exhibition design

G4_USB LiDAR





JINGMENG

(Strange Dreamland)

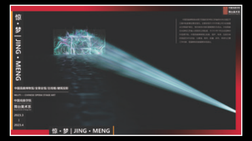
Other Works	07
Theme	Intangible Cultural Heritage
Work type	3D mapping （Architectural Projection）
Video link	*https://youtu.be/9Xl-65xvT-c*

Brief Introduction

The protection of intangible cultural heritage is currently one of the most important research topics. Kunqu Opera is one of the first batch of national intangible cultural heritage. The term 'Jingmeng' comes from Kunqu Opera, meaning "a strange dreamland", and this work is also a creative inheritance, protection and research of Kunqu Opera, showcasing it in a contemporary form.

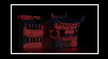
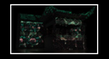
Suzhou kunqu Museum 3dmapping projection and research on the production of irregular projections at the national grand theatre

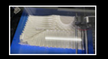
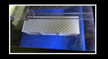
Poster design for 'Jing Meng'



3D Printing

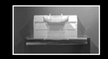
3D Mapping

Spray painting and drying

Splicing and bonding



Exhibition at the National Grand Theatre

Creative team: NACTA Department of Stage Design - Qijing Visual New Media Art Studio

Project manager: Wang Tingting，Feng Ziqian

Digital production: Feng Ziqian

Type of work: 3D mapping projection

Dimensions of the work: 88 * 48 * 46 cm (60 : 1)

The ancient theatre hall 'China Kunqu Opera Museum', situated within the Quanjin Guild Hall on Zhangjia Lane, Pingjiang Road in Suzhou's historic city centre, stands as a nationally designated key cultural relic protection unit. It houses Suzhou's most exquisite ancient Kunqu opera stage.

Dedicated to rescuing, preserving, transmitting and promoting the ancient art of Kunqu opera, architectural projection performances serve as an excellent medium for this purpose. A miniature model of the venue was exhibited at the The National Grand Theater.

Suzhou kunqu museum 3D mapping projection storyboard